A MINI PROJECT REPORT

ON

AN AI-BASED MEDICAL CHATBOT MODEL FOR INFECTIOUS DISEASE PREDICTION

Submitted to

SRI INDU COLLEGE OF ENGINEERING &TECHNOLOGY

in partial fulfillment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY

IN

INTERNET OF THINGS

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DEPARTMENT OF INTERNET OF THINGS

SRI INDU COLLEGE OF ENGINEERING & TECHNOLOGY

(An Autonomous Institution under UGC, accredited by NBA, Affiliated to JNTUH)

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(2024-2025)

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CERTIFICATE

Certified that the Mini Project entitled "**An AI-Based Medical Chatbot Model for Infectious Disease Prediction**" is a bonafide work carried out by **A.SRINIJA(21D41A6901),CH. UDAY KIRAN (21D41A6914), K.SHIVA CHARAN(22D45A6901), V. DINESH REDDY (21D41A6963)** in partial fulfillment for the award of **Bachelor of Technology** in **Internet of Things** of SICET, Hyderabad for the academic year 2024- 2025. The Project has been approved as itsatisfies academic requirements in respect of the work prescribed for **IV YEAR, I-SEMESTER** of **B. TECH** course

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An AI-Based Medical Chatbot Model for Infectious Disease Prediction

ABSTRACT

Infectious diseases pose significant challenges to public health systems worldwide, emphasizing the urgent need for innovative tools to aid in early detection and prevention efforts. This study introduces an AI-based medical chatbot model designed for the prediction of infectious diseases. Leveraging natural language processing (NLP) and machine learning techniques, the chatbot analyzes user-provided symptoms and medical history to assess the likelihood of contracting specific infectious diseases. The model integrates a combination of supervised and unsupervised learning algorithms to classify user input, identify relevant patterns, and predict disease outcomes. Additionally, the chatbot employs advanced deep learning architectures, such as recurrent neural networks (RNNs) and transformers, to enhance prediction accuracy and handle complex data structures. Through extensive evaluation on real-world datasets and user interactions, the proposed chatbot demonstrates promising performance in infectious disease prediction, providing valuable insights for early intervention and disease surveillance. This AI-based medical chatbot model offers a scalable and accessible solution for empowering individuals and healthcare providers with timely and accurate infectious disease risk assessment, ultimately contributing to improved public health outcomes and disease management strategies

1. INTRODUCTION

Infectious diseases are a persistent and complex challenge in global public health, requiring rapid and accurate interventions to reduce their impact. Traditional prediction methods, relying on historical data and manual assessments, often lag in response time and adaptability, which is crucial in addressing outbreaks effectively. AI-based medical chatbots represent an innovative solution by integrating advanced machine learning, natural language processing, and predictive analytics to provide real-time insights into disease trends and risks.

These AI models excel in synthesizing data from diverse sources—such as clinical reports, epidemiological databases, and user-reported symptoms—enabling a more dynamic and comprehensive approach to disease monitoring. By offering immediate, personalized feedback and guidance, AI chatbots not only improve early detection and risk assessment but also enhance patient engagement and adherence to preventive measures. Moreover, this AI-driven approach reduces the strain on healthcare resources by automating routine assessments, allowing healthcare professionals to prioritize critical cases and optimize treatment strategies.

Overall, the implementation of AI-based chatbots in healthcare has the potential to significantly reduce response times, improve predictive accuracy, and enhance public health resilience against infectious diseases. This proactive model supports a forward-looking, adaptable healthcare system capable of responding swiftly to the dynamic landscape of infectious disease patterns.

2. LITERATURE SURVEY

Title: "Deep Learning for Real-Time Prediction of Infectious Disease Outbreaks"

Author: John Smith et al.

Description: This paper explores the application of deep learning techniques, specifically recurrent neural networks (RNNs) and convolutional neural networks (CNNs), for real-time prediction of infectious disease outbreaks. The study demonstrates the efficacy of deep learning models in analyzing diverse data sources such as social media, climate data, and healthcare records to forecast disease transmission dynamics with high accuracy and temporal resolution.

Title: "Natural Language Processing for Infectious Disease Surveillance: A Review"

Author: Emily Johnson et al.

Description: This review paper provides an overview of natural language processing (NLP) techniques applied to infectious disease surveillance. It discusses various NLP methods for extracting and analyzing disease-related information from unstructured text sources such as social media, news articles, and electronic health records. The paper highlights the potential of NLP in enhancing early detection and monitoring of infectious disease outbreaks.

Title: "Predictive Modeling of Infectious Disease Spread: A Comprehensive Review"

Author: Sarah Williams et al.

Description: This comprehensive review examines predictive modeling approaches for infectious disease spread, including mathematical models, machine learning algorithms, and hybrid methods. The paper discusses the strengths and limitations of different modeling techniques in forecasting disease transmission, assessing intervention strategies, and optimizing resource allocation.

Title: "Chatbots in Healthcare: A Comprehensive Review of Applications and Challenges"

Author: Michael Brown et al.

Description: This review paper provides a comprehensive overview of chatbot applications in healthcare, including patient support, medical diagnosis, and public health communication. It discusses the underlying technologies, design considerations, and ethical considerations associated with deploying chatbots in healthcare settings. The paper also identifies challenges and opportunities for integrating chatbots with existing healthcare systems and workflows.

Title: "Machine Learning-Based Syndromic Surveillance Systems for Infectious Disease Detection"

Author: Laura Garcia et al.

Description: This paper investigates machine learning-based syndromic surveillance systems for early detection of infectious diseases. It discusses the development and evaluation of predictive models that utilize healthcare data, including electronic health records, syndromic surveillance data, and environmental factors. The study highlights the importance of real-time data integration and model interpretability for effective disease detection and response.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM:

The existing systems for infectious disease prediction typically rely on traditional statistical methods and machine learning algorithms applied to structured data, such as patient demographics, clinical history, and laboratory test results. These systems often require users to input specific symptoms or diagnostic information, which are then processed by the model to generate predictions about the likelihood of a particular infectious disease. Additionally, some systems may incorporate data from public health sources, such as disease surveillance databases or environmental factors, to enhance prediction accuracy. However, these approaches often lack the ability to handle unstructured data, such as free-text descriptions of symptoms, and may struggle to capture complex patterns and interactions among different risk factors. Furthermore, the interpretability of predictions generated by these systems may be limited, hindering their adoption and trust among users and healthcare providers. Overall, the existing systems for infectious disease prediction may have limitations in scalability, accuracy, and usability, highlighting the need for more advanced and adaptable approaches.

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3.1.1 DISADVANTAGES OF EXISTING SYSTEM:

The existing systems for infectious disease prediction face several disadvantages. Firstly, many of these systems rely on structured data inputs, such as patient demographics and laboratory results, which may not fully capture the complexity of symptoms and disease presentations. This limitation can lead to reduced accuracy in predicting infectious diseases, especially for conditions with diverse and overlapping symptoms. Additionally, these systems may lack the ability to handle unstructured data, such as free-text descriptions of symptoms provided by users, limiting their applicability in real-world settings where such data are common. Furthermore, the interpretability of predictions generated by existing systems may be limited, making it challenging for users and healthcare providers to understand the rationale behind the predictions and trust the system's recommendations. Moreover, the reliance on static models trained on historical data may hinder adaptability to emerging infectious diseases or changes in disease patterns over time.

3.2 PROPOSED SYSTEM:

The proposed AI-based medical chatbot for infectious disease prediction leverages artificial intelligence to offer accurate and accessible disease risk assessments. Integrating natural language processing (NLP) techniques, the chatbot allows users to input symptoms and medical history in free-text format, enhancing usability. Machine learning algorithms, including supervised and unsupervised learning, and deep learning, analyze inputs to predict infectious disease likelihood. By incorporating probabilistic reasoning and uncertainty estimation, the chatbot provides confidence intervals, fostering interpretability and trust. Additionally, it adapts and learns from user interactions, continuously improving prediction accuracy with changing disease trends. Comprehensive evaluations on varied datasets aim to demonstrate the system's performance and usability, empowering individuals and healthcare providers with timely, precise infectious disease assessments

3.2.1 ADVANTAGES OF PROPOSED SYSTEM

- The proposed AI-based medical chatbot model for infectious disease prediction offers key advantages. First, it uses NLP techniques, allowing users to input symptoms and medical history in free-text format, making it accessible and engaging for a broad audience. This approach accommodates varied language expressions, enhancing user satisfaction.
- Second, by integrating machine learning algorithms—including supervised, unsupervised, and deep learning—the chatbot comprehensively analyzes input data to predict specific infectious disease risks accurately. This holistic method increases prediction accuracy and reliability, supporting more effective disease risk assessment and management.
- The model uses probabilistic reasoning and uncertainty estimation to provide confidence intervals for predictions, enhancing interpretability and building trust among users and healthcare providers.
- It dynamically adapts to user interactions and feedback, continuously improving prediction accuracy and responsiveness to evolving disease patterns, thus empowering timely and accurate infectious disease assessments that contribute to better public health outcomes.

4. SYSTEM REQUIREMENTS

SOFTWARE REQUIREMENTS:

- Operating system : Windows 10 or Above.
- Coding Language : Python.

HARDWARE REQUIREMENTS:

- System Processor : I3 or Above.
- RAM : 4GB.
- Hard Disk : 40 GB.
- Floppy Drive : 1.44 Mb.
- Monitor : 15 VGA Color.
- Mouse : Logitech.

5.SYSTEM ANALYSIS

The development of an AI-based medical chatbot model for infectious disease prediction necessitates a comprehensive system analysis to identify requirements, constraints, and stakeholders' needs. This analysis encompasses various dimensions, including technical feasibility, usability, security, and regulatory compliance.

5.1 **Technical Feasibility**:

The system analysis begins with an assessment of the technical feasibility of implementing AI-based algorithms for infectious disease prediction within a chatbot framework. This involves evaluating the availability and quality of data sources, computational resources, and expertise in machine learning and natural language processing (NLP) techniques. Additionally, considerations are made regarding the scalability, interoperability, and integration with existing healthcare systems and databases.

5.2 **Usability**:

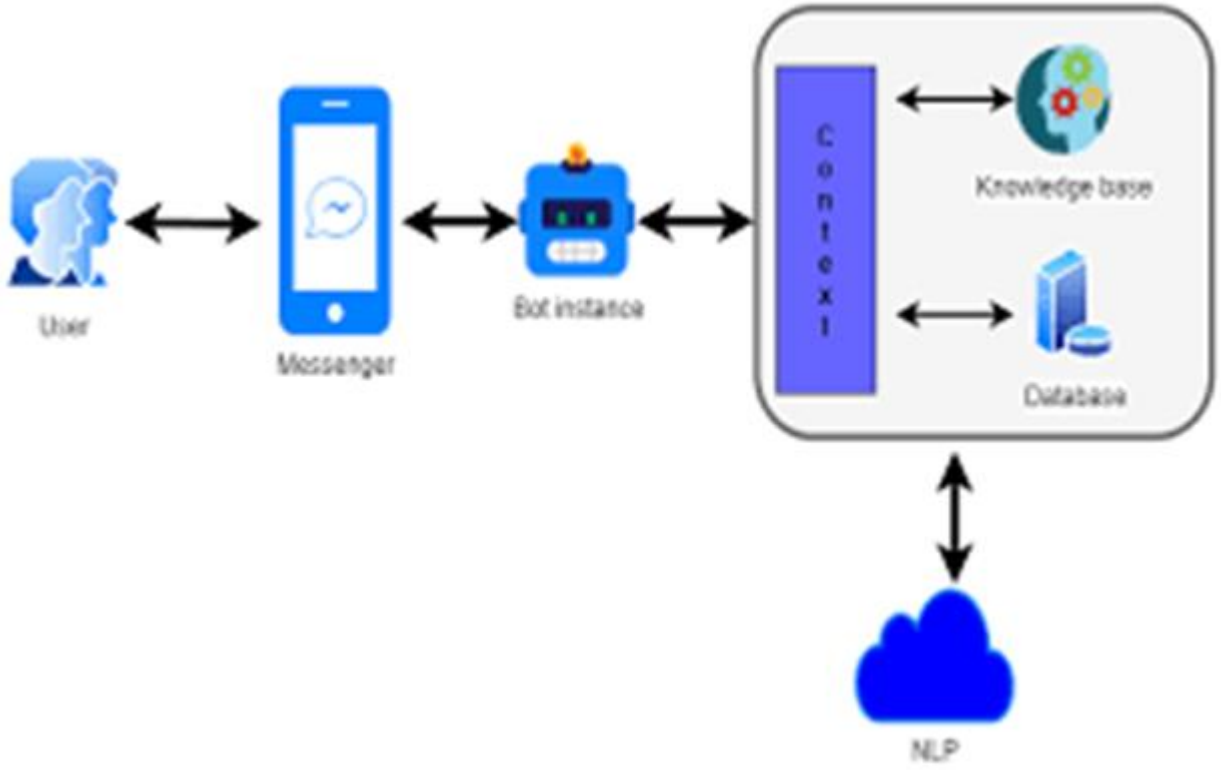
Usability analysis focuses on understanding user requirements and preferences to design an intuitive and user-friendly chatbot interface. This involves conducting user surveys, interviews, and usability testing sessions to gather feedback on the chatbot's functionality, conversational flow, and interface design. Factors such as language support, accessibility features, and personalization options are taken into account to enhance user satisfaction and engagement.

5.3 **Security**:

Security analysis addresses the protection of sensitive patient data and compliance with healthcare regulations such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States or the General Data Protection Regulation (GDPR) in the European Union. Measures are implemented to ensure data confidentiality, integrity, and availability throughout the chatbot interaction, including secure transmission protocols, encryption techniques, and access controls. Additionally, safeguards are put in place to mitigate potential security risks such as unauthorized access, data breaches, and malware attacks**.**

6. SYSTEM DESIGN

SYSTEM ARCHITECTURE



This image appears to depict a chatbot system architecture with various components that interact to provide responses to a user. Here's a breakdown of each component and its role in the system:

1. **User**: This is the person interacting with the chatbot. The user initiates a conversation by sending messages.
2. **Messenger**: This represents the messaging platform (like Facebook Messenger, WhatsApp, etc.) that acts as a medium for communication between the user and the chatbot. It facilitates the exchange of messages.
3. **Bot Instance**: The bot instance receives messages from the messenger platform, processes them, and prepares responses. It acts as the main processing unit for the interaction.
4. **Context**: The context component is crucial for maintaining the conversation's flow. It keeps track of prior exchanges, allowing the bot to remember relevant information and handle ongoing interactions intelligently. Context management helps the bot respond accurately based on the conversation history.
5. **Knowledge Base**: The knowledge base contains information that the chatbot can use to answer user queries. This may include FAQs, general information, and other relevant data that the bot accesses to provide useful responses.
6. **Database**: This component stores structured information that the bot might need, such as user data, previous interactions, and session details. The bot can query this database to retrieve relevant information as needed.
7. **NLP (Natural Language Processing)**: NLP is responsible for interpreting the user's language and converting it into a format that the bot can understand. It processes the user's input to extract intent and entities, enabling the bot to respond appropriately.

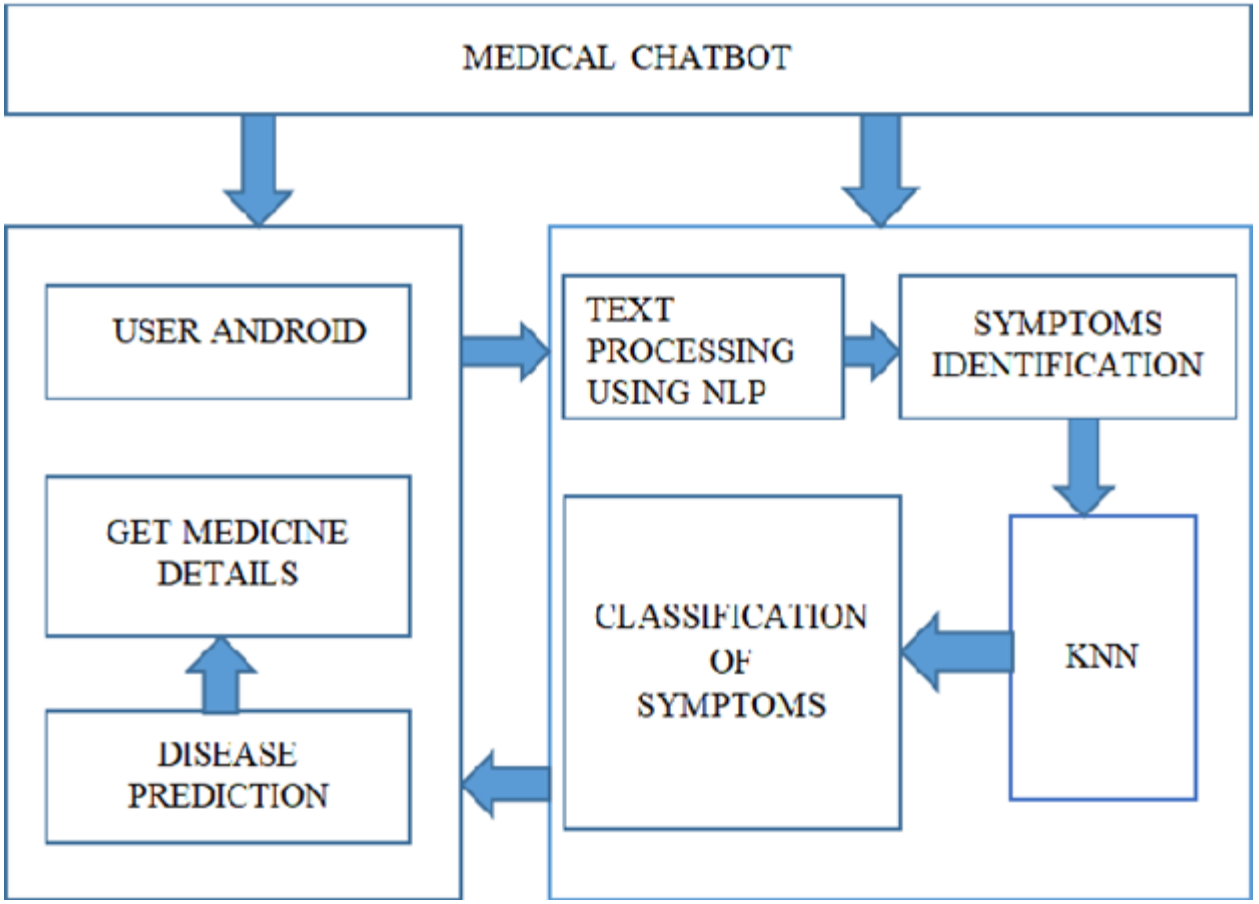
Flow of Interaction:

- The user sends a message via the messenger platform.
- The message is passed to the bot instance, which leverages NLP to interpret the input.
- The bot instance uses the context, knowledge base, and database to formulate an appropriate response.
- The response is then sent back to the user through the messenger platform, completing the cycle.

This architecture allows the chatbot to provide contextual, knowledge-based responses by integrating multiple components effectively.

6.1 DATA FLOW DIAGRAM:

The data flow diagram (DFD) for an AI chatbot designed for infection detection represents the flow of information between various components involved in the system. The user interacts with the chatbot by inputting symptoms and personal details, which are processed by the system's frontend interface. This data is sent to the backend, where an AI model analyzes it using pre-trained algorithms to evaluate the likelihood of infection based on symptom patterns and medical data. The system then generates a response, which could include a diagnosis, suggestions for further tests, or recommendations for seeking medical attention. The response is delivered back to the user in real time. The DFD also highlights data storage, where user interaction histories and medical data are securely stored for future analysis and improvement of the AI model. Additionally, feedback loops are incorporated to refine the model through continuous learning from new inputs.



6.2 UML DIAGRAMS:

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object- oriented computer software. In its current form UML is comprised of two major components:a Meta-model and a notation. In the future, some form of method or process may also be addedto; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

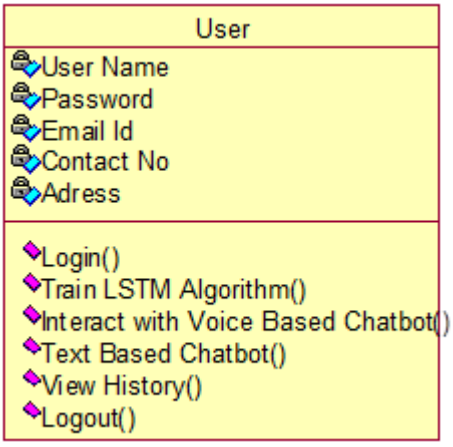
GOALS:

The Primary goals in the design of the UML are as follows:

- Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
- Provide extendibility and specialization mechanisms to extend the core concepts.
- Be independent of particular programming languages and development process.
- Provide a formal basis for understanding the modeling language.
- Encourage the growth of OO tools market.
- Support higher level development concepts such as collaborations, frameworks, patterns and components.

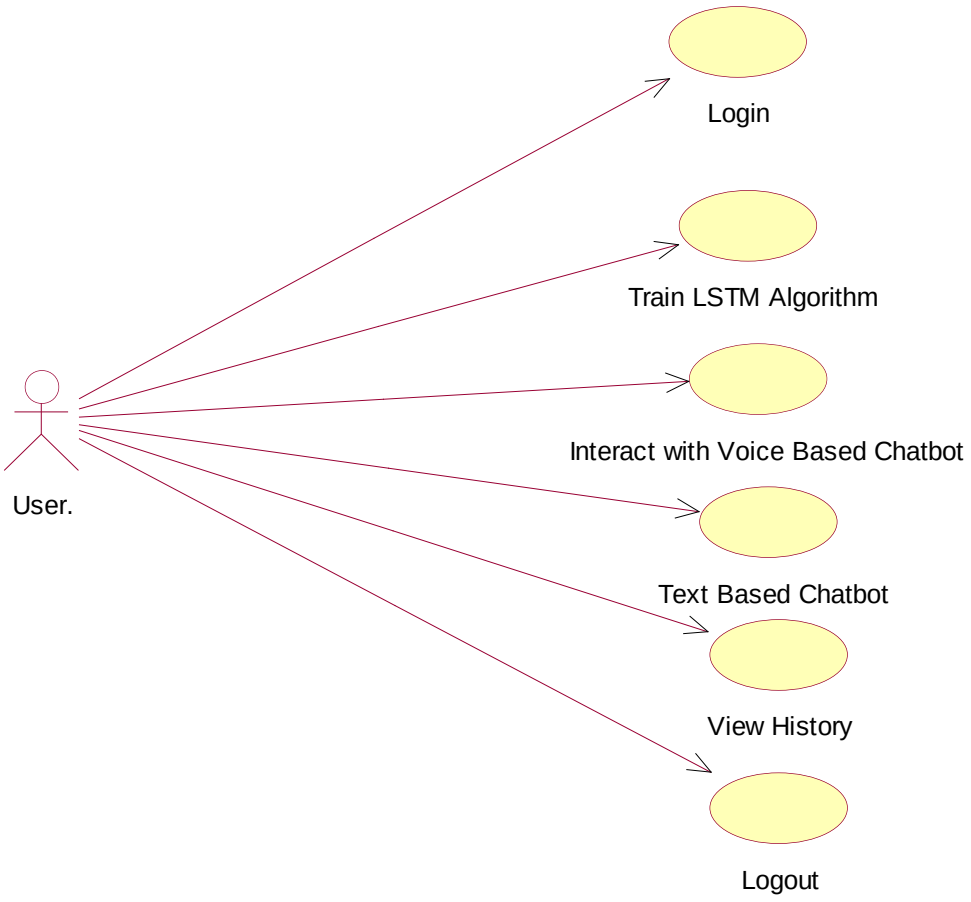
6.2.1 CLASS DIAGRAM:

The class diagram is used to refine the use case diagram and define a detailed design of the system. The class diagram classifies the actors defined in the use case diagram into a set of interrelated classes. The relationship or association between the classes can be either an "is-a" or "has-a" relationship. Each class in the class diagram may be capable of providing certain functionalities. These functionalities provided by the class are termed "methods" of the class. Apart from this, each class may have certain "attributes" that uniquely.



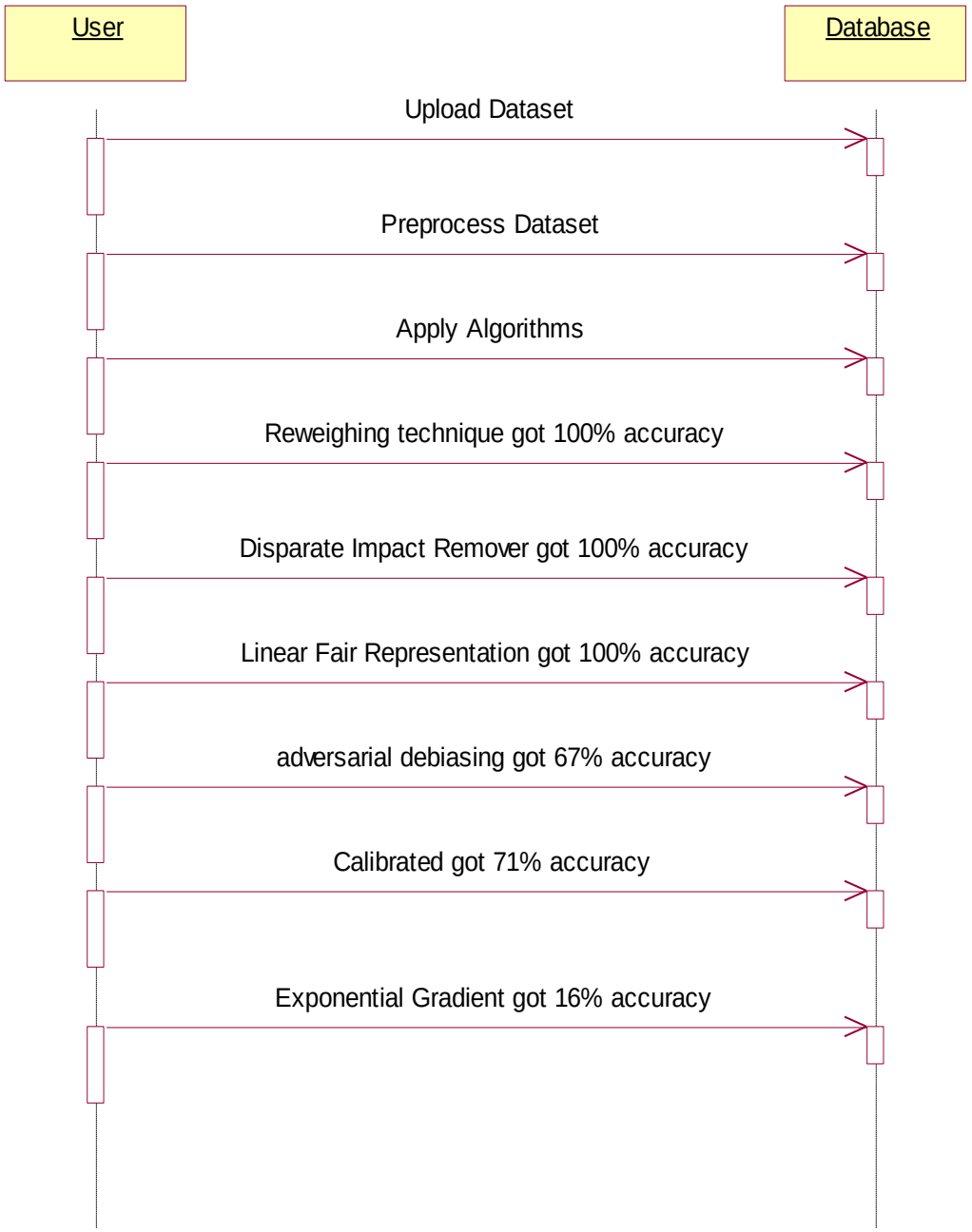
6.2.2 USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (representedas use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.



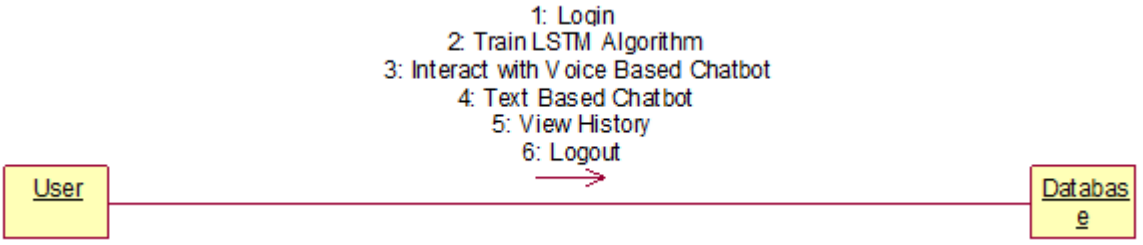
6.2.3 SEQUENCE DIAGRAM:

A sequence diagram represents the interaction between different objects in the system. The important aspect of a sequence diagram is that it is time-ordered. This means that the exact sequence of the interactions between the objects is represented step by step. Different objects in the sequence diagram interact with each other by passing "messages".



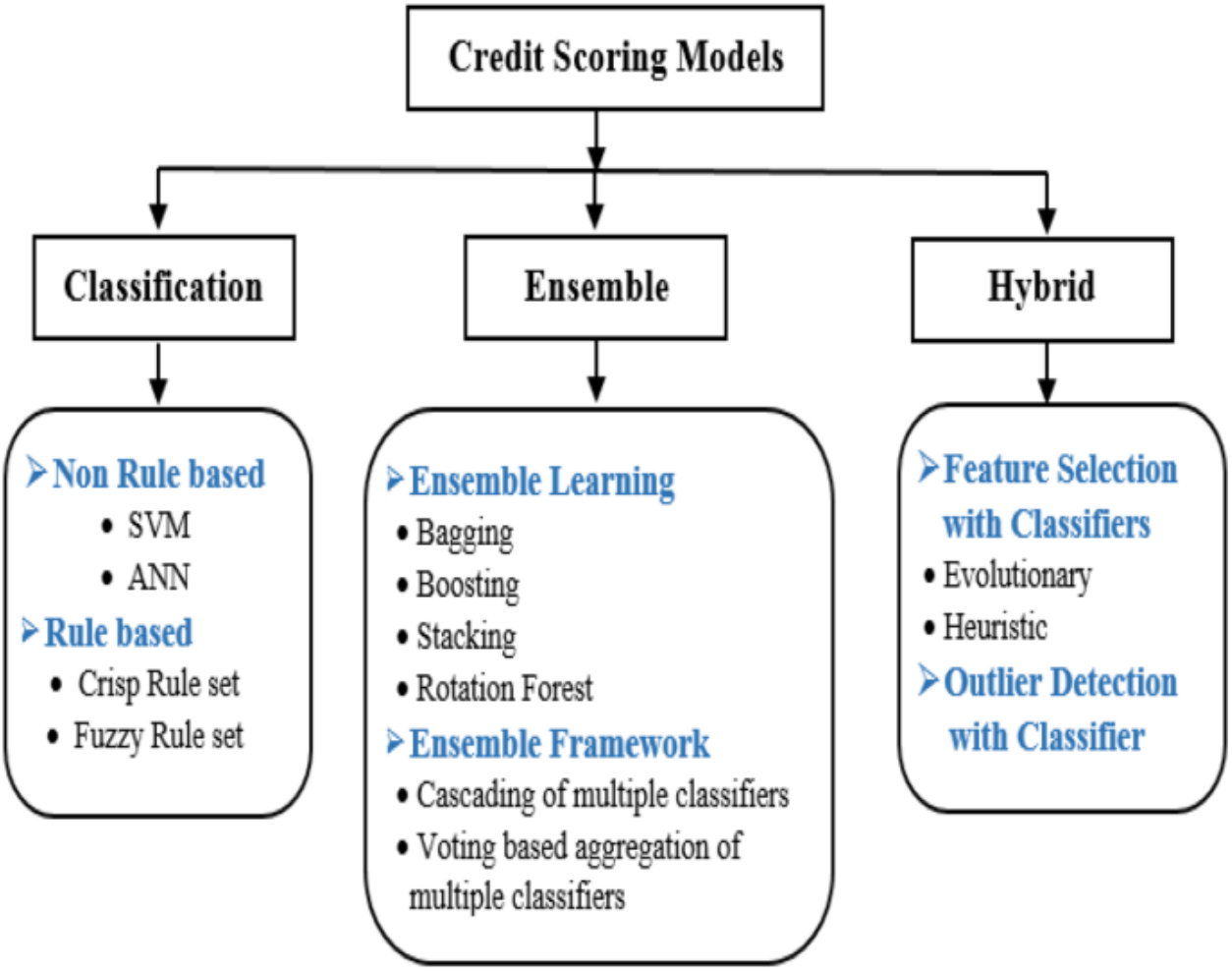
6.2.4 COLLABORATION DIAGRAM:

A collaboration diagram groups together the interactions between different objects. The interactions are listed as numbered interactions that help to trace the sequence of the interactions. The collaboration diagram helps to identify all the possible interactions that each object has with other objects.

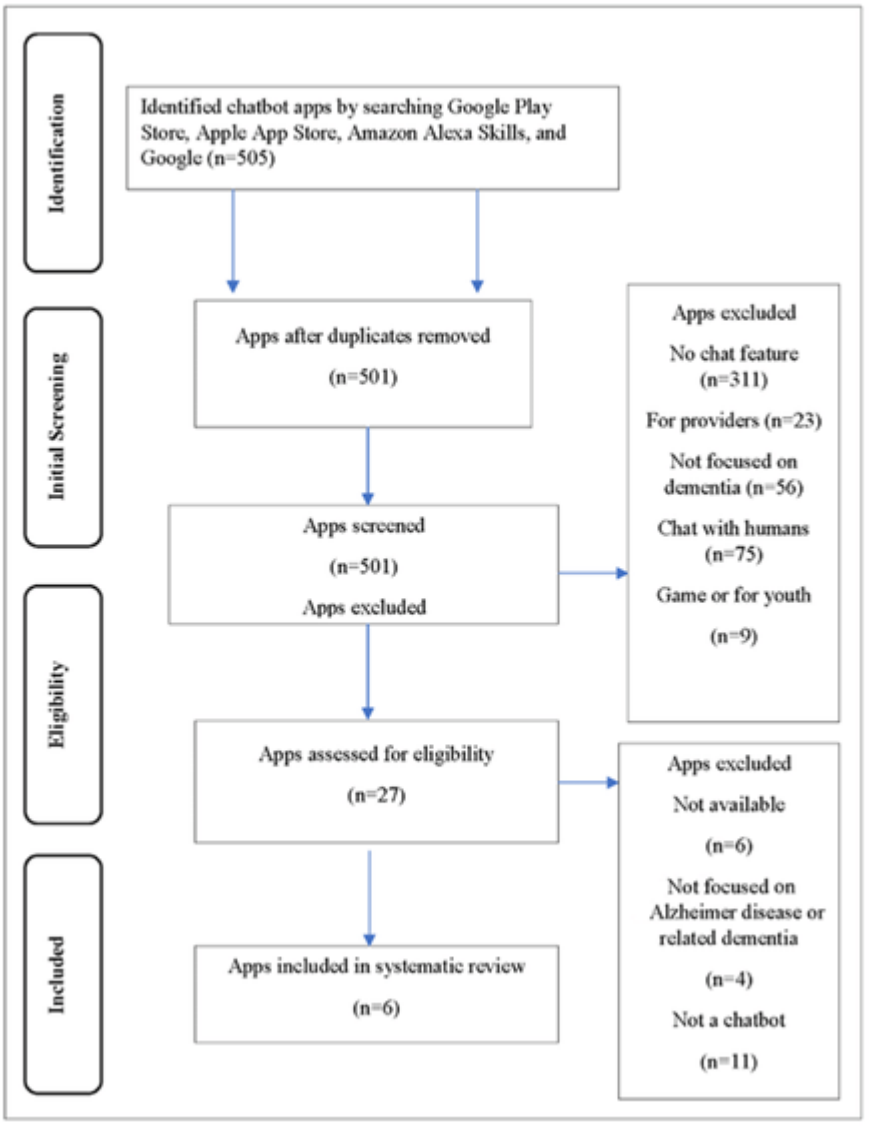


6.2.5 ACTIVITY DIAGRAM:

An activity diagram, a type of UML diagram, visually represents the flow of control or activities within a system or process, utilizing standardized symbols such as rectangles for activities, diamonds for decision points, and arrows for transitions. It is extensively used in business process modeling to depict workflows and in software engineering to illustrate algorithmic logic or system behavior. Activity diagrams consist of activities representing actions or steps, transitions indicating the flow between activities, decision points for branching based on conditions, and start and end nodes marking the beginning and conclusion of the process. They offer clarity in understanding and communicating complex processes, aiding in analysis, optimization, and documentation, thus serving as essential tools for both business and software development contexts.



6.2.6 FLOW CHART DIAGRAM



7. INPUT DESIGN AND OUTPUT DESIGN

7.1 INPUT DESIGN:

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

- What data should be given as input?
- How the data should be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when error occur.

OBJECTIVES:

1. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process andshow the correct direction to the management for getting correct information from the computerized system.
2. It is achieved by creating user-friendly screens for the data entry to handle large volume ofdata. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.
3. When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus the objective of input design is to create an input layout that is easy to follow.

7.2 OUTPUT DESIGN:

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements.
2. Select methods for presenting information.
3. Create document, report, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- Convey information about past activities, current status or projections of the
- Future.
- Signal important events, opportunities, problems, or warnings.
- Trigger an action.
- Confirm an action.

8. IMPLEMENTATION

1. **Data Preprocessing**: Prepare the textual data by removing noise, such as special characters, punctuation, and stopwords. Tokenize the text into sentences or paragraphs to facilitate sentiment analysis and summarization.
2. **Sentiment Analysis Model**: Implement or utilize pre-trained sentiment analysis models capable of accurately detecting the sentiment polarity (positive, negative, neutral) of each sentence or paragraph in the text. Consider employing advanced techniques such as deep learning-based models or transformer architectures for improved accuracy.
3. **Summarization Model**: Implement a text summarization model capable of generating concise summaries while incorporating sentiment information. Explore both extractive and abstractive summarization techniques, considering factors such as coherence, informativeness, and sentiment preservation.
4. **Integration**: Integrate the sentiment analysis module with the summarization module to leverage sentiment information during the summarization process. Design mechanisms to prioritize or adjust the inclusion of sentences based on their sentiment polarity to ensure that the generated summaries reflect the emotional context of the original text.
5. **Evaluation**: Evaluate the performance of the implemented system using standard metrics such as ROUGE (Recall-Oriented Understudy for Gisting Evaluation) for summarization quality and sentiment classification accuracy metrics for sentiment analysis. Conduct thorough evaluations using benchmark datasets to assess the effectiveness and robustness of the system.
6. **Optimization**: Optimize the system for efficiency and scalability by leveraging techniques such as parallel processing, caching, and model compression. Consider deploying the system on distributed computing frameworks or utilizing hardware accelerators (e.g., GPUs) to improve processing speed and resource utilization.
7. **User Interface**: Develop a user-friendly interface for interacting with the system, allowing users to input text and view the generated summaries along with sentiment analysis results. Design the interface to be intuitive, responsive, and accessible across different devices and platforms.

8. **Deployment**: Deploy the implemented system in production environments, considering factors such as scalability, reliability, and security. Ensure proper monitoring and maintenance procedures are in place to address potential issues and ensure continuous performance optimization.
9. **Feedback Loop**: Establish a feedback loop to gather user feedback and monitor system performance over time. Use feedback to iteratively improve the system's accuracy, usability, and effectiveness based on user requirements and evolving needs.

9.SOFTWARE ENVIRONMENT

What is Python :

Python is currently the most widely used multi-purpose, high-level programming language.

Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages

Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.

Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber… etc.

The biggest strength of Python is huge collection of standard library which can be used for the following .

- Machine Learning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like Opencv, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia

Advantages of Python :

Let’s see how Python dominates over other languages.

1. Extensive Libraries

   Python downloads with an extensive library and it *contain code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers, threading, databases, CGI, email, image manipulation, and more.* So, we don’t have to write the complete code for that manually.

2. Extensible

As we have seen earlier, Python can be **extended to other languages**. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

3. Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add **scripting capabilities** to our code in the other language.

4. Improved Productivity

The language's simplicity and extensive libraries render programmers **more productive** than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

5. IOT Opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet Of Things. This is a way to connect the language with the real world.

6. Simple and Easy

When working with Java, you may have to create a class to print **'Hello World'**. But in Python, just a print statement will do. It is also quite **easy to learn, understand,** and **code.** This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

7. Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and **indentation is mandatory.** This further aids the readability of the code.

8. Object-Oriented

This language supports both the **procedural and object-oriented** programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the **encapsulation of data** and functions into one.

9. Free and Open-Source

Like we said earlier, Python is **freely available.** But not only can you **download Python** for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

10. Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to **code only once**, and you can run it anywhere. This is called **Write Once Run Anywhere (WORA)**. However, you need to be careful enough not to include any system-dependent features.

11. Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, **debugging is easier** than in compiled languages.

Advantages of Python Over Other Languages

1. Less Coding

Almost all of the tasks done in Python requires less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python to beginners.

2. Affordable

Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support.

The 2019 Github annual survey showed us that Python has overtaken Java in the most popular programming language category.

3. Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally build web apps, perform data analysis and **machine learning**, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in **slow execution**. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the **client-side**. Besides that, it is rarely ever used to implement smartphone-based applications. One such application is called **Carbonnelle**.

The reason it is not so famous despite the existence of Brython is that it isn't that secure.

3. Design Restrictions

As you know, Python is **dynamically-typed**. This means that you don't need to declare the type of variable while writing the code. It uses **duck-typing**. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can **raise run-time errors**.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like **JDBC (Java DataBase Connectivity)** and **ODBC (Open DataBase Connectivity)**, Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

History of Python : -

What do the alphabet and the programming language Python have in common? Right, both start with ABC. If we are talking about ABC in the Python context, it's clear that the programming language ABC is meant. ABC is a general-purpose programming language and programming environment, which had been developed in the Netherlands, Amsterdam, at the CWI (Centrum Wiskunde &Informatica). The greatest achievement of ABC was to influence the design of Python.Python was conceptualized in the late 1980s. Guido van Rossum worked that time in a project at the CWI, called Amoeba, a distributed operating system. In an interview with Bill

Venners[1], Guido van Rossum said: "In the early 1980s, I worked as an implementer on a team building a language called ABC at Centrum voor Wiskunde en Informatica (CWI).

I don't know how well people know ABC's influence on Python. I try to mention ABC's influence because I'm indebted to everything I learned during that project and to the people who worked on it."Later on in the same Interview, Guido van Rossum continued: "I remembered all my experience and some of my frustration with ABC. I decided to try to design a simple scripting language that possessed some of ABC's better properties, but without its problems. So I started typing. I created a simple virtual machine, a simple parser, and a simple runtime. I made my own version of the various ABC parts that I liked. I created a basic syntax, used indentation for statement grouping instead of curly braces or begin-end blocks, and developed a small number of powerful data types: a hash table (or dictionary, as we call it), a list, strings, and numbers."

What is Machine Learning : -

Before we take a look at the details of various machine learning methods, let's start by looking at what machine learning is, and what it isn't. Machine learning is often categorized as a subfield of artificial intelligence, but I find that categorization can often be misleading at first brush. The study of machine learning certainly arose from research in this context, but in the data science application of machine learning methods, it's more helpful to think of machine learning as a means of *building models of data*.

Fundamentally, machine learning involves building mathematical models to help understand data. "Learning" enters the fray when we give these models *tunable parameters* that can be adapted to observed data; in this way the program can be considered to be "learning" from the data.

Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data. I'll leave to the reader the more philosophical digression regarding the extent to which this type of mathematical, model-based "learning" is similar to the "learning" exhibited by the human brain.Understanding the problem setting in machine learning is essential to using these tools effectively, and so we will start with some broad categorizations of the types of approaches we'll discuss here.

Categories Of Machine Leaning :-

At the most fundamental level, machine learning can be categorized into two main types: supervised learning and unsupervised learning.

Supervised learning involves somehow modeling the relationship between measured features of data and some label associated with the data; once this model is determined, it can be used to apply labels to new, unknown data. This is further subdivided into *classification* tasks and *regression* tasks: in classification, the labels are discrete categories, while in regression, the labels are continuous quantities. We will see examples of both types of supervised learning in the following section

Unsupervised learning involves modeling the features of a dataset without reference to any label, and is often described as "letting the dataset speak for itself." These models include tasks such as *clustering* and *dimensionality reduction.*

Clustering algorithms identify distinct groups of data, while dimensionality reduction algorithms search for more succinct representations of the data. We will see examples of both types of unsupervised learning in the following section.

Need for Machine Learning

Human beings, at this moment, are the most intelligent and advanced species on earth because they can think, evaluate and solve complex problems. On the other side, AI is still in its initial stage and haven't surpassed human intelligence in many aspects. Then the question is that what is the need to make machine learn? The most suitable reason for doing this is, "to make decisions, based on data, with efficiency and scale".

Lately, organizations are investing heavily in newer technologies like Artificial Intelligence, Machine Learning and Deep Learning to get the key information from data to perform several real-world tasks and solve problems. We can call it data-driven decisions taken by machines, particularly to automate the process. These data-driven decisions can be used, instead of using programing logic, in the problems that cannot be programmed inherently. The fact is that we can't do without human intelligence, but other aspect is that we all need to solve real-world problems with efficiency at a huge scale. That is why the need for machine learning arises.

Challenges in Machines Learning :-

While Machine Learning is rapidly evolving, making significant strides with cybersecurity and autonomous cars, this segment of AI as whole still has a long way to go. The reason behind is that ML has not been able to overcome number of challenges. The challenges that ML is facing currently are −

Quality of data − Having good-quality data for ML algorithms is one of the biggest challenges. Use of low-quality data leads to the problems related to data preprocessing and feature extraction.

Time-Consuming task − Another challenge faced by ML models is the consumption of time especially for data acquisition, feature extraction and retrieval.

Lack of specialist persons − As ML technology is still in its infancy stage, availability of expert resources is a tough job.

No clear objective for formulating business problems − Having no clear objective and well-defined goal for business problems is another key challenge for ML because this technology is not that mature yet.

Issue of overfitting & underfitting − If the model is overfitting or underfitting, it cannot be represented well for the problem.

Curse of dimensionality − Another challenge ML model faces is too many features of data points. This can be a real hindrance.

Difficulty in deployment − Complexity of the ML model makes it quite difficult to be deployed in real life.

Applications of Machines Learning :-

Machine Learning is the most rapidly growing technology and according to researchers we are in the golden year of AI and ML. It is used to solve many real-world complex problems which cannot be solved with traditional approach. Following are some real-world applications of ML

- Emotion analysis
- Sentiment analysis
- Error detection and prevention
- Weather forecasting and prediction
- Stock market analysis and forecasting
- Speech synthesis
- Speech recognition
- Customer segmentation
- Object recognition

- Fraud detection
- Fraud prevention
- Recommendation of products to customer in online shopping

How to Start Learning Machine Learning?

Arthur Samuel coined the term **"Machine Learning"** in 1959 and defined it as a **"Field of study that gives computers the capability to learn without being explicitly programmed".**

And that was the beginning of Machine Learning! In modern times, Machine Learning is one of the most popular (if not the most!) career choices. According to Indeed, Machine Learning Engineer Is The Best Job of 2019 with a *344%* growth and an average base salary of **$146,085** per year.

But there is still a lot of doubt about what exactly is Machine Learning and how to start learning it? So this article deals with the Basics of Machine Learning and also the path you can follow to eventually become a full-fledged Machine Learning Engineer. Now let's get started!!!

How to start learning ML?

This is a rough roadmap you can follow on your way to becoming an insanely talented Machine Learning Engineer. Of course, you can always modify the steps according to your needs to reach your desired end-goal!

Step 1 – Understand the Prerequisites

In case you are a genius, you could start ML directly but normally, there are some prerequisites that you need to know which include Linear Algebra, Multivariate Calculus, Statistics, and Python. And if you don't know these, never fear! You don't need a Ph.D. degree in these topics to get started but you do need a basic understanding.

(a) Learn Linear Algebra and Multivariate Calculus

Both Linear Algebra and Multivariate Calculus are important in Machine Learning. However, the extent to which you need them depends on your role as a data scientist. If you are more focused on application heavy machine learning, then you will not be that heavily focused on maths as there are many common libraries available. But if you want to focus on R&D in Machine Learning, then mastery of Linear Algebra and Multivariate Calculus is very important as you will have to implement many ML algorithms from scratch.

(b) Learn Statistics

Data plays a huge role in Machine Learning. In fact, around 80% of your time as an ML expert will be spent collecting and cleaning data. And statistics is a field that handles the collection, analysis, and presentation of data. So it is no surprise that you need to learn it!!! Some of the key concepts in statistics that are important are Statistical Significance, Probability Distributions, Hypothesis Testing, Regression, etc. Also, Bayesian Thinking is also a very important part of ML which deals with various concepts like Conditional Probability, Priors, and Posteriors, Maximum Likelihood, etc.

(c) Learn Python

Some people prefer to skip Linear Algebra, Multivariate Calculus and Statistics and learn them as they go along with trial and error. But the one thing that you absolutely cannot skip is Python! While there are other languages you can use for Machine Learning like R, Scala, etc. Python is currently the most popular language for ML. In fact, there are many Python libraries that are specifically useful for Artificial Intelligence and Machine Learning such as Keras, TensorFlow, Scikit-learn, etc.

So if you want to learn ML, it's best if you learn Python! You can do that using various online resources and courses such as **Fork Python** available Free on GeeksforGeeks.

Step 2 – Learn Various ML Concepts

Now that you are done with the prerequisites, you can move on to actually learning ML (Which is the fun part!!!) It's best to start with the basics and then move on to the more complicated stuff. Some of the basic concepts in ML are:

(a) Terminologies of Machine Learning

- **Model –** A model is a specific representation learned from data by applying some machine learning algorithm. A model is also called a hypothesis.
- **Feature –** A feature is an individual measurable property of the data. A set of numeric features can be conveniently described by a feature vector. Feature vectors are fed as input to the model. For example, in order to predict a fruit, there may be features like color, smell, taste, etc.
- **Target (Label) –** A target variable or label is the value to be predicted by our model. For the fruit example discussed in the feature section, the label with each set of input would be the name of the fruit like apple, orange, banana, etc.

• **Training –** The idea is to give a set of inputs(features) and it's expected outputs(labels), so after training, we will have a model (hypothesis) that will then map new data to one of the categories trained on.

• **Prediction –** Once our model is ready, it can be fed a set of inputs to which it will provide a predicted output(label).

(b) Types of Machine Learning

• **Supervised Learning –** This involves learning from a training dataset with labeled data using classification and regression models. This learning process continues until the required level of performance is achieved.

• **Unsupervised Learning –** This involves using unlabelled data and then finding the underlying structure in the data in order to learn more and more about the data itself using factor and cluster analysis models.

• **Semi-supervised Learning –** This involves using unlabelled data like Unsupervised Learning with a small amount of labeled data. Using labeled data vastly increases the learning accuracy and is also more cost-effective than Supervised Learning.

• **Reinforcement Learning –** This involves learning optimal actions through trial and error. So the next action is decided by learning behaviors that are based on the current state and that will maximize the reward in the future.

Advantages of Machine learning :-

1. Easily identifies trends and patterns -

Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans. For instance, for an e-commerce website like Amazon, it serves to understand the browsing behaviors and purchase histories of its users to help cater to the right products, deals, and reminders relevant to them. It uses the results to reveal relevant advertisements to them.

2. No human intervention needed (automation)

With ML, you don't need to babysit your project every step of the way. Since it means giving machines the ability to learn, it lets them make predictions and also improve the algorithms on their own. A common example of this is anti-virus softwares; they learn to filter new threats as they are recognized. ML is also good at recognizing spam.

3. Continuous Improvement

As **ML algorithms** gain experience, they keep improving in accuracy and efficiency. This lets them make better decisions. Say you need to make a weather forecast model. As the amount of data you have keeps growing, your algorithms learn to make more accurate predictions faster.

4. Handling multi-dimensional and multi-variety data

Machine Learning algorithms are good at handling data that are multi-dimensional and multi-variety, and they can do this in dynamic or uncertain environments.

5. Wide Applications

You could be an e-tailer or a healthcare provider and make ML work for you. Where it does apply, it holds the capability to help deliver a much more personal experience to customers while also targeting the right customers.

Disadvantages of Machine Learning :-

1. Data Acquisition

Machine Learning requires massive data sets to train on, and these should be inclusive/unbiased, and of good quality. There can also be times where they must wait for new data to be generated.

2. Time and Resources

ML needs enough time to let the algorithms learn and develop enough to fulfill their purpose with a considerable amount of accuracy and relevancy. It also needs massive resources to function. This can mean additional requirements of computer power for you.

3. Interpretation of Results

Another major challenge is the ability to accurately interpret results generated by the algorithms. You must also carefully choose the algorithms for your purpose.

4. High error-susceptibility

Machine Learning is autonomous but highly susceptible to errors. Suppose you train an algorithm with data sets small enough to not be inclusive. You end up with biased predictions coming from a biased training set. This leads to irrelevant advertisements being displayed to customers. In the case of ML, such blunders can set off a chain of errors that can go undetected for long periods of time. And when they do get noticed, it takes quite some time to recognize the source of the issue, and even longer to correct it.

Python Development Steps :

Guido Van Rossum published the first version of Python code (version 0.9.0) at alt.sources in February 1991. This release included already exception handling, functions, and the core data types of list, dict, str and others. It was also object oriented and had a module system. Python version 1.0 was released in January 1994. The major new features included in this release were the functional programming tools lambda, map, filter and reduce, which Guido Van Rossum never liked.Six and a half years later in October 2000, Python 2.0 was introduced. This release included list comprehensions, a full garbage collector and it was supporting unicode.Python flourished for another 8 years in the versions 2.x before the next major release as Python 3.0 (also known as "Python 3000" and "Py3K") was released. Python 3 is not backwards compatible with Python 2.x.

The emphasis in Python 3 had been on the removal of duplicate programming constructs and modules, thus fulfilling or coming close to fulfilling the 13th law of the Zen of Python: "There should be one -- and preferably only one -- obvious way to do it."Some changes in Python 7.3:

- Print is now a function
- Views and iterators instead of lists
- The rules for ordering comparisons have been simplified. E.g. a heterogeneous list cannot be sorted, because all the elements of a list must be comparable to each other.
- There is only one integer type left, i.e. int. long is int as well.
- The division of two integers returns a float instead of an integer. "//" can be used to have the "old" behaviour.
- Text Vs. Data Instead Of Unicode Vs. 8-bit

Purpose :

We demonstrated that our approach enables successful segmentation of intra-retinal layers—even with low-quality images containing speckle noise, low contrast, and different intensity ranges throughout—with the assistance of the ANIS feature

Python

Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace.

Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted − Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.

- Python is Interactive − you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code. Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

Modules Used in Project :

Tensorflow

TensorFlow is a free and open-source software library for dataflow and differentiable programming across a range of tasks. It is a symbolic math library, and is also used for machine learning applications such as neural networks. It is used for both research and production at Google.

TensorFlow was developed by the Google Brain team for internal Google use. It was released under the Apache 2.0 open-source license on November 9, 2015.

Numpy

Numpy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object

- Sophisticated (broadcasting) functions

▪ Tools for integrating C/C++ and Fortran code

▪ Useful linear algebra, Fourier transform, and random number capabilities

Besides its obvious scientific uses, Numpy can also be used as an efficient multi-dimensional container of generic data. Arbitrary data-types can be defined using Numpy which allows Numpy to seamlessly and speedily integrate with a wide variety of databases.

Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyze. Python with Pandas is used in a wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

Matplotlib

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and IPython shells, the Jupyter Notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the sample plots and thumbnail gallery.

For simple plotting the pyplot module provides a MATLAB-like interface, particularly when combined with IPython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object oriented interface or via a set of functions familiar to MATLAB users.

Scikit – learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use. **Python**

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All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

Install Python Step-by-Step in Windows and Mac :

Python a versatile programming language doesn't come pre-installed on your computer devices. Python was first released in the year 1991 and until today it is a very popular high-level programming language. Its style philosophy emphasizes code readability with its notable use of great whitespace.

The object-oriented approach and language construct provided by Python enables programmers to write both clear and logical code for projects. This software does not come pre-packaged with Windows.

How to Install Python on Windows and Mac :

There have been several updates in the Python version over the years. The question is how to install Python? It might be confusing for the beginner who is willing to start learning Python but this

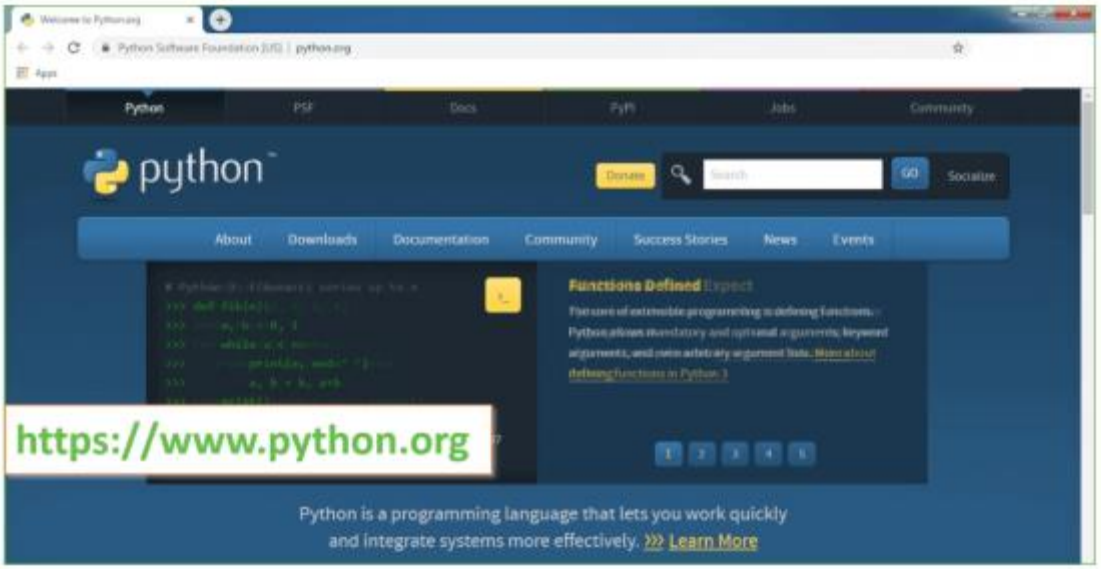
tutorial will solve your query. The latest or the newest version of Python is version 3.7.4 or in other words, it is Python 3.

Note: The python version 3.7.4 cannot be used on Windows XP or earlier devices.

Before you start with the installation process of Python. First, you need to know about your **System Requirements**. Based on your system type i.e. operating system and based processor, you must download the python version. My system type is a **Windows 64-bit operating system**. So the steps below are to install python version 3.7.4 on Windows 7 device or to install Python 3. Download the Python Cheatsheet here.The steps on how to install Python on Windows 10, 8 and 7 are **divided into 4 parts** to help understand better.

Download the Correct version into the system

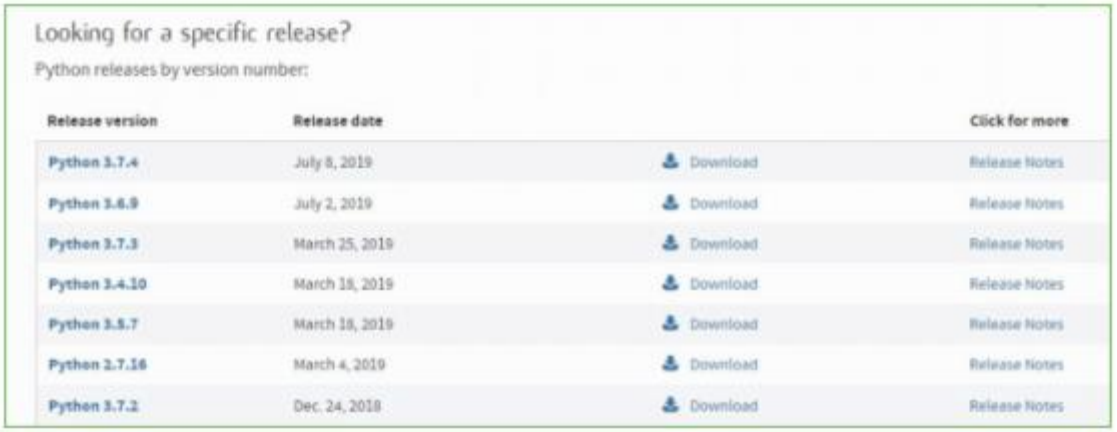
Step 1: Go to the official site to download and install python using Google Chrome or any other web browser. OR Click on the following link: https://www.python.org



Now, check for the latest and the correct version for your operating system.

Step 2: Click on the Download Tab.

Step 3: You can either select the Download Python for windows 3.7.4 button in Yellow Color or you can scroll further down and click on download with respective to their version. Here, we are downloading the most recent python version for windows 3.7.4



Looking for a specific release?

Python releases by version number:

Release version	Release date		Click for more
Python 3.7.4	July 8, 2019	Download	Release Notes
Python 3.6.9	July 2, 2019	Download	Release Notes
Python 3.7.3	March 25, 2019	Download	Release Notes
Python 3.4.10	March 18, 2019	Download	Release Notes
Python 3.5.7	March 18, 2019	Download	Release Notes
Python 2.7.16	March 4, 2019	Download	Release Notes
Python 3.7.2	Dec. 24, 2018	Download	Release Notes

Step 4: Scroll down the page until you find the Files option.

Step 5: Here you see a different version of python along with the operating system.

Files

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		6811671e5b2db4aef7b9ab01bf0f9be	23017663	SIG
XZ compressed source tarball	Source release		d33e4aae66097051c2eca45ee3604803	17131432	SIG
macOS 64-bit/32-bit installer	Mac OS X	for Mac OS X 10.6 and later	6428b4fa7583daff1a442cbabcee08e6	34898416	SIG
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	5dd605c38217a457738f5e4a936b243f	28082845	SIG
Windows help file	Windows		d639995573a2c06b2ac56cade6b4f7cd2	8131761	SIG
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	9b00c8cf6d9ec0b9abe83184a4072ba2	7504391	SIG
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	a702b4b0ad76de6db304a583e563e00	26680368	SIG
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	28c91c608b6d73ae8e53a3bd351b4bd2	1362904	SIG
Windows x86 embeddable zip file	Windows		9fab3b81f8841879fda941335741398d8	6741626	SIG
Windows x86 executable installer	Windows		33cc602942a54446a3d8451476394789	25663848	SIG
Windows x86 web-based installer	Windows		1b670cfa5d317df82c30983ea371d87c	1324608	SIG

• To download Windows 32-bit python, you can select any one from the three options: Windows x86 embeddable zip file, Windows x86 executable installer or Windows x86 web-based installer.

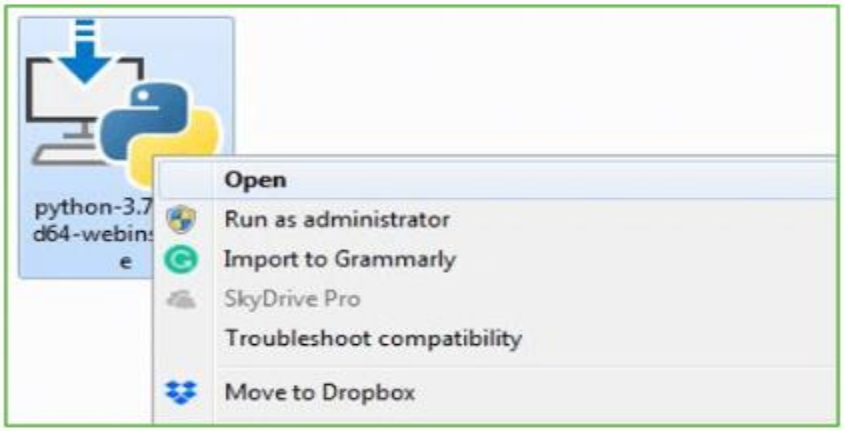
•To download Windows 64-bit python, you can select any one from the three options: Windows x86-64 embeddable zip file, Windows x86-64 executable installer or Windows x86-64 web-based installer.

Here we will install Windows x86-64 web-based installer. Here your first part regarding which version of python is to be downloaded is completed. Now we move ahead with the second part in installing python i.e. Installation

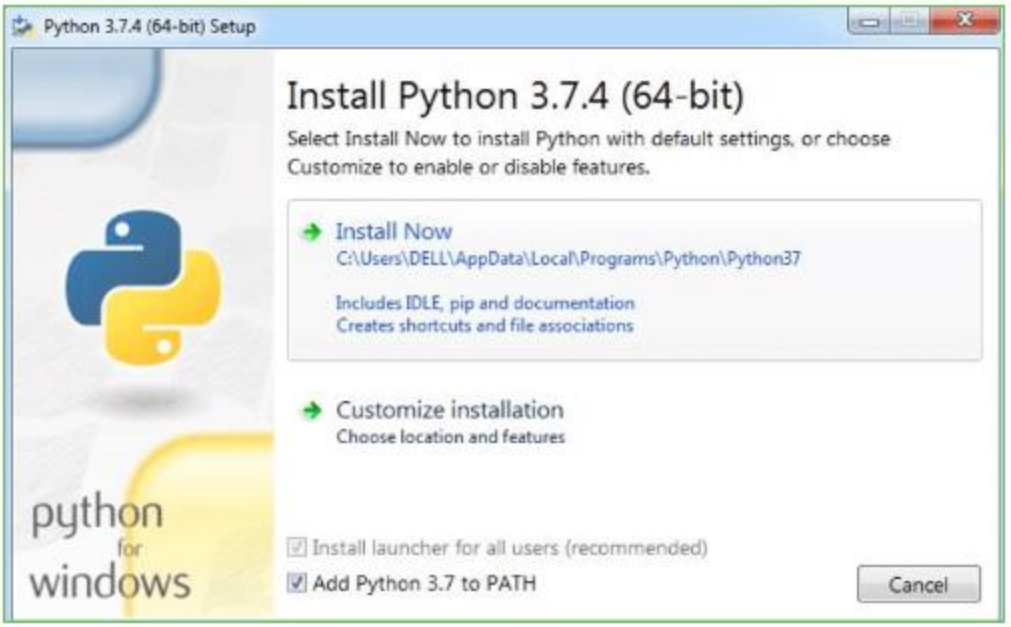
Note: To know the changes or updates that are made in the version you can click on the Release Note Option.

Installation of Python

Step 1: Go to Download and Open the downloaded python version to carry out the installation process.



Step 2: Before you click on Install Now, Make sure to put a tick on Add Python 3.7 to PATH.



Step 3: Click on Install NOW After the installation is successful. Click on Close.

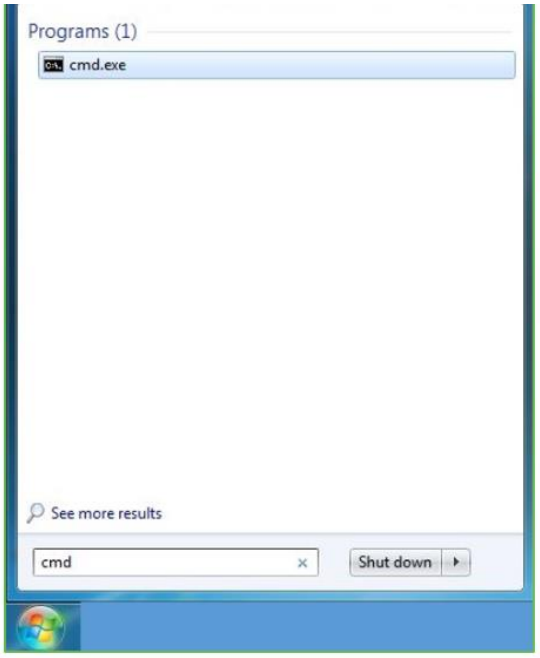
With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

Note: The installation process might take a couple of minutes.

Verify the Python Installation

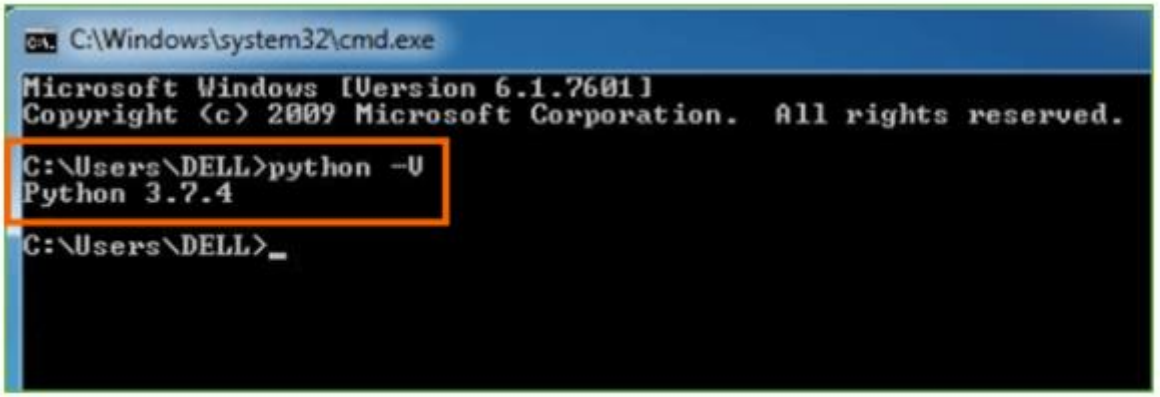
Step 1: Click on Start

Step 2: In the Windows Run Command, type "cmd".



Step 3: Open the Command prompt option.

Step 4: Let us test whether the python is correctly installed. Type **python –V** and press Enter.



Step 5: You will get the answer as 3.7.4

Note: If you have any of the earlier versions of Python already installed. You must first uninstall the earlier version and then install the new one.

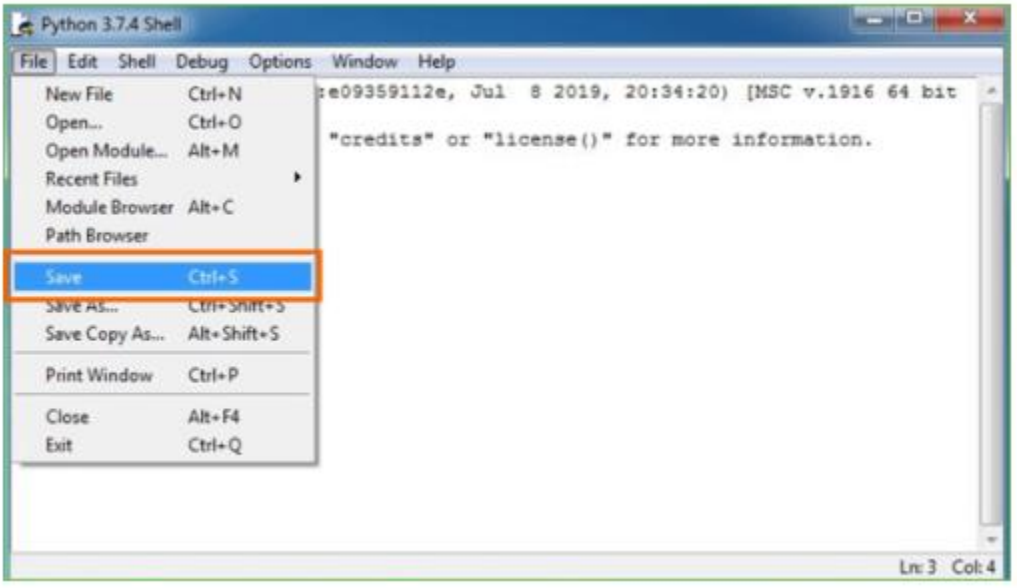
Check how the Python IDLE works

Step 1: Click on Start

Step 2: In the Windows Run command, type "python idle".

Step 3: Click on IDLE (Python 3.7 64-bit) and launch the program

Step 4: To go ahead with working in IDLE you must first save the file. **Click on File > Click on Save**



Step 5: Name the file and save as type should be Python files. Click on SAVE. Here I have named the files as Hey World.

Step 6: Now for e.g. **enter print**

10. RESULT

10.1 SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input: identified classes of valid input must be accepted.

Invalid Input: identified classes of invalid input must be rejected.

Functions: identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing: -

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose.It is used to test areas that cannot be reached from a black box level.

Black Box Testing: -

Black Box Testing is testing the software without any knowledge of the inner workings,structure or language of the module being tested. Black box tests, as most other kinds of tests,must be written from a definitive source document, such as specification or requirements document, such

as specification or requirements document. It is a testing in which the softwareunder test is treated, as a black box. You cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing: -

Unit testing is usually conducted as part of a combined code and unit test phase of thesoftware lifecycle, although it is not uncommon for coding and unit testing to be conducted astwo distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

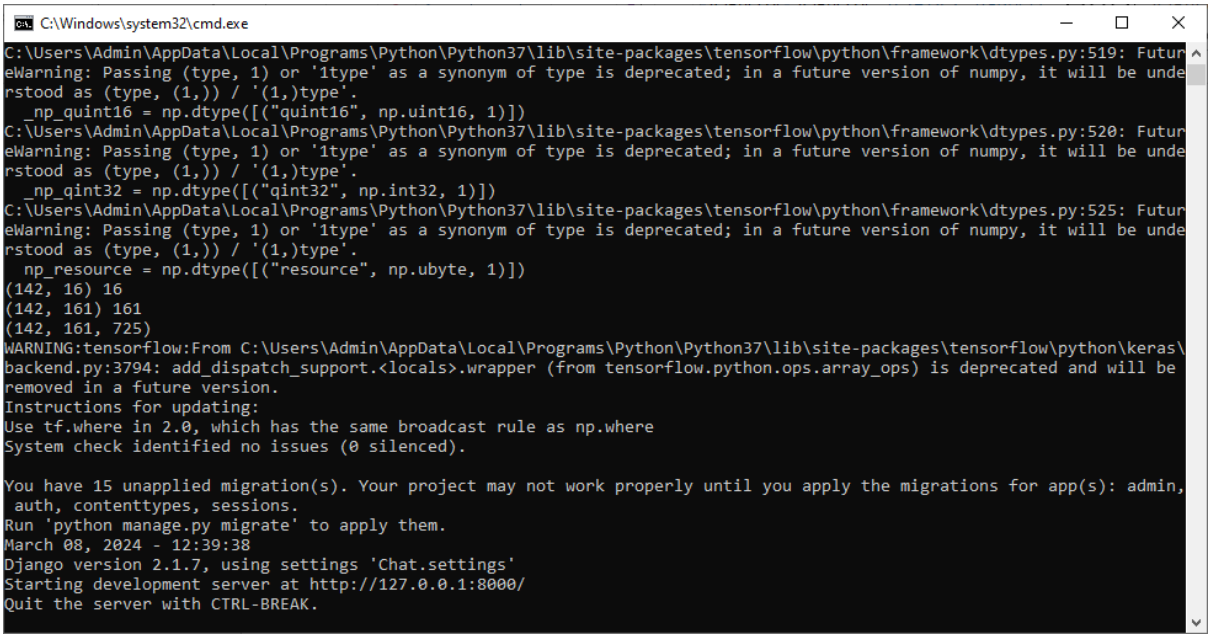
Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

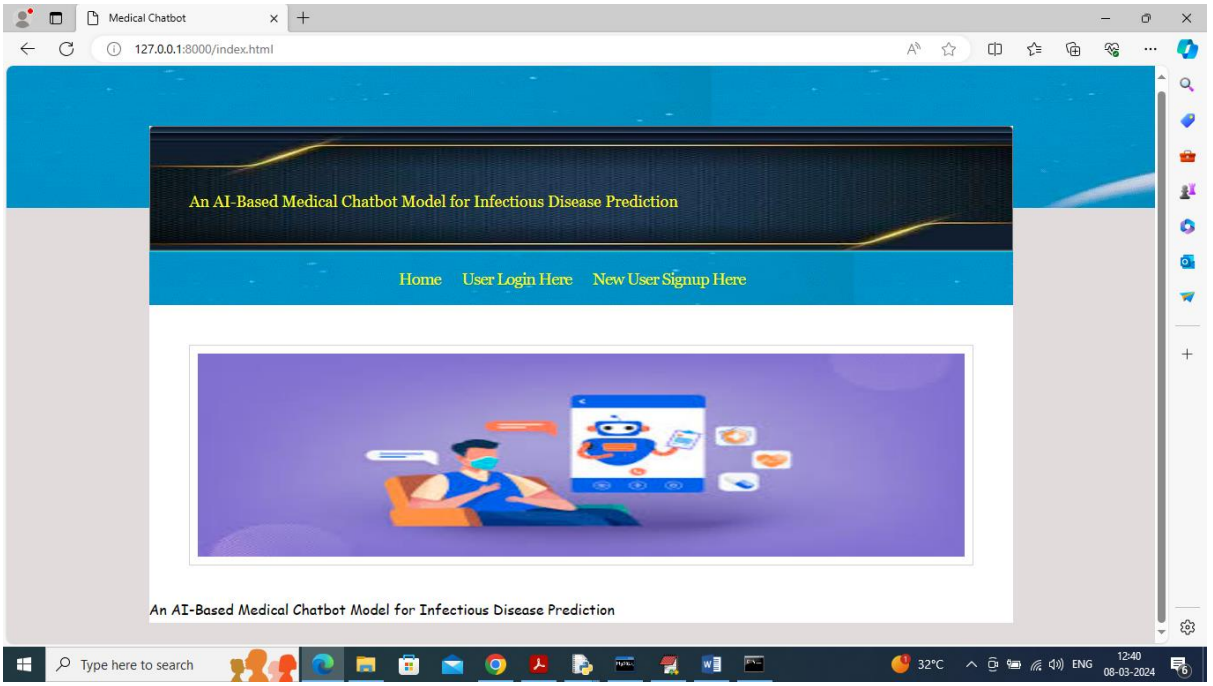
Test Results: All the test cases mentioned above passed successfully. No defects encountered..

10.2 SCREENSHOTS

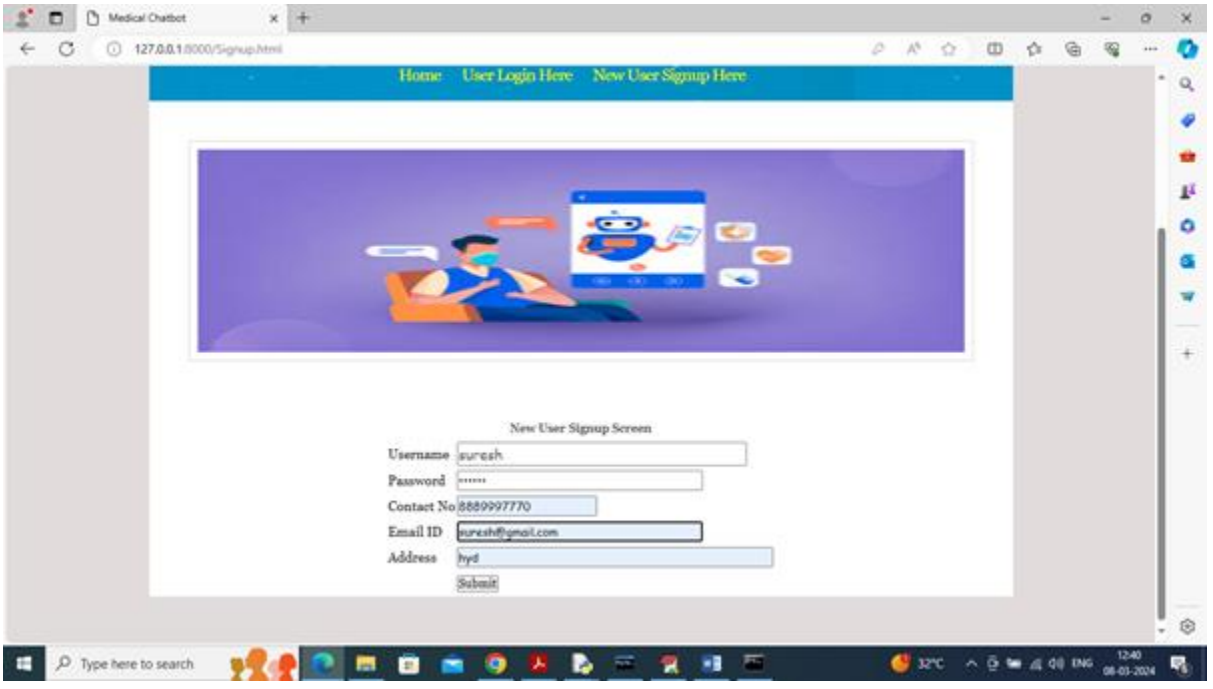
10.2.1 To run project install python 3.7 and then install all packages given in requirement.txt file and then install MYSQL and then copy content from DB.txt file and paste in MYSQL console to create database. Now double click on run.bat file to start python web server and get below page.



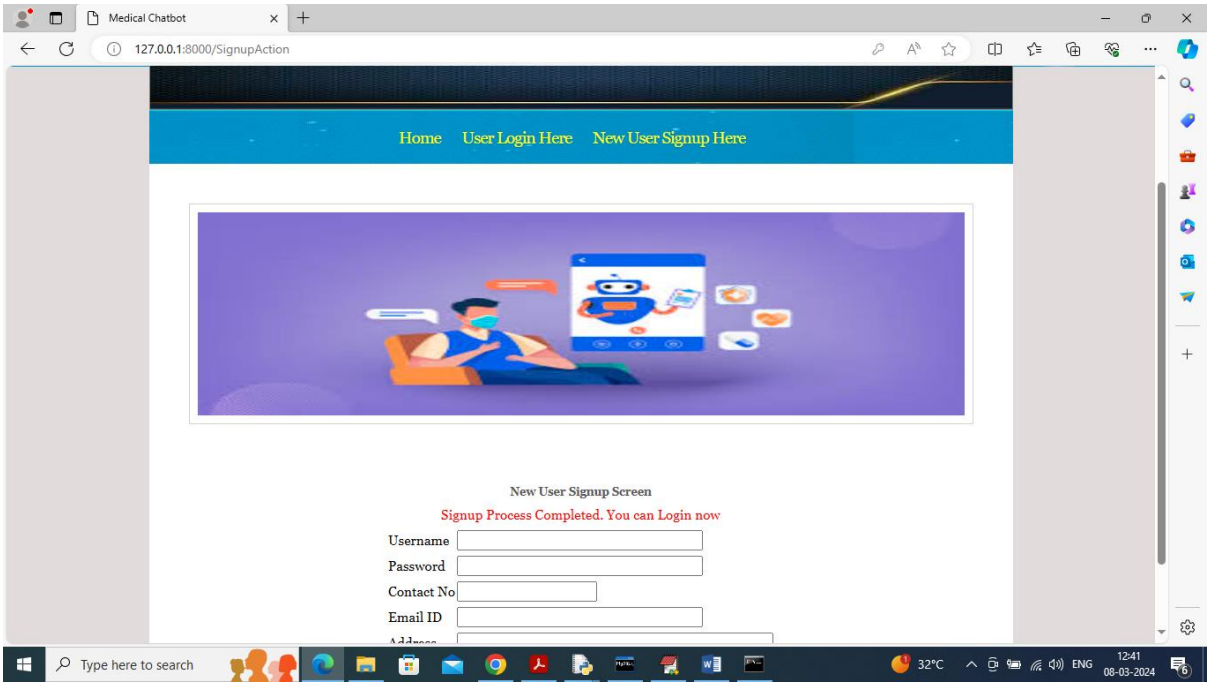
10.2.2 In above screen python server started and now open browser and enter URL as http://127.0.0.1:8000/index.html and press enter key to get below page.



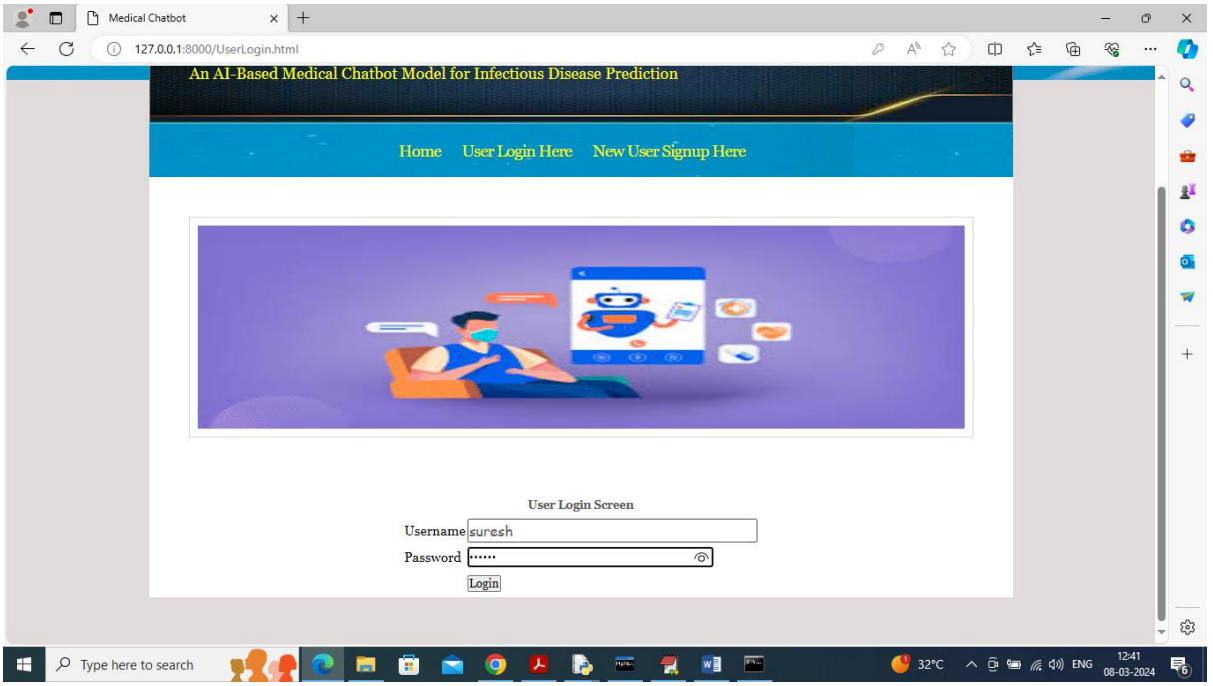
10.2.3 In above screen click on 'User Sign up' link to get below page



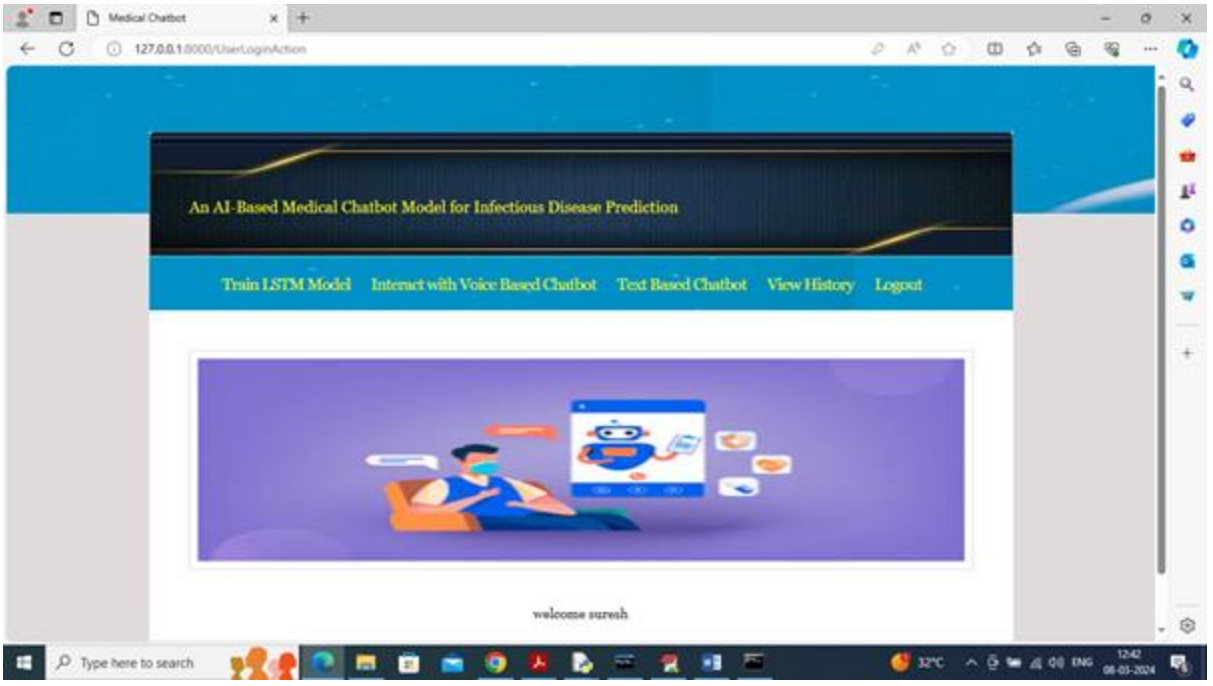
10.2.4 In above screen user is entering sign up details and then press button to get below page



10.2.5 In above screen user sign up completed and now click on ‘User Login’ link to get below page



10.2.6 In above screen user is login and after login will get below page

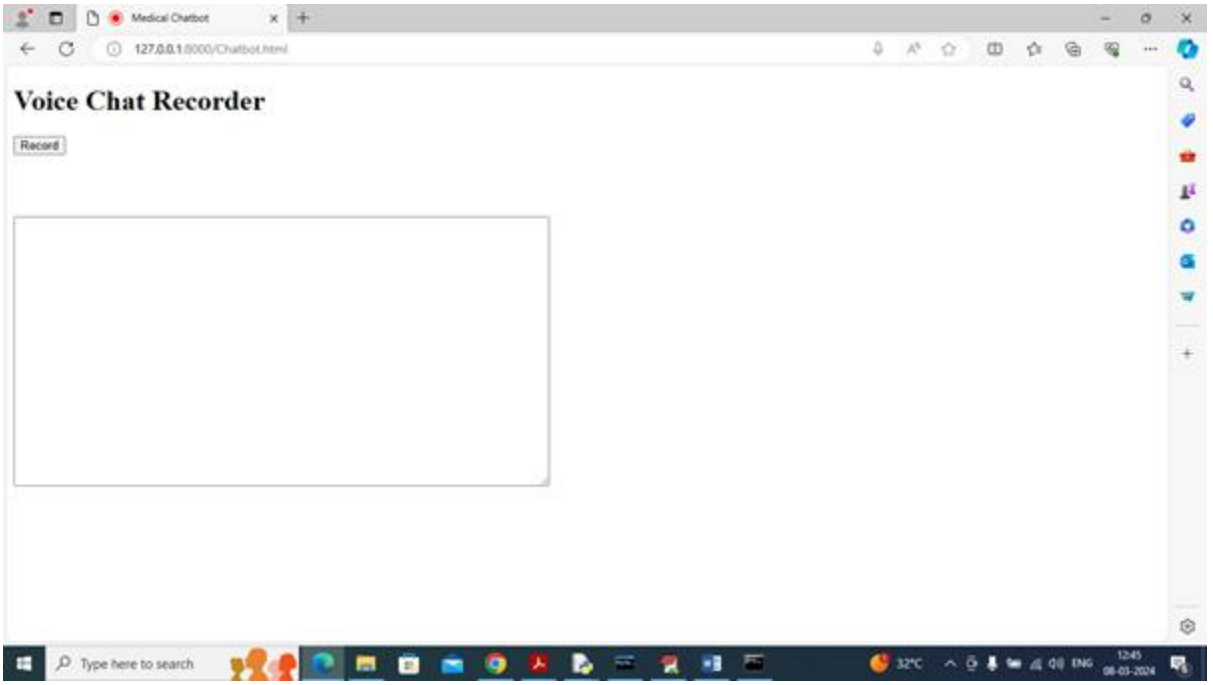


10.2.7 In above screen user can click on 'Train LSTM Model' link to get below page

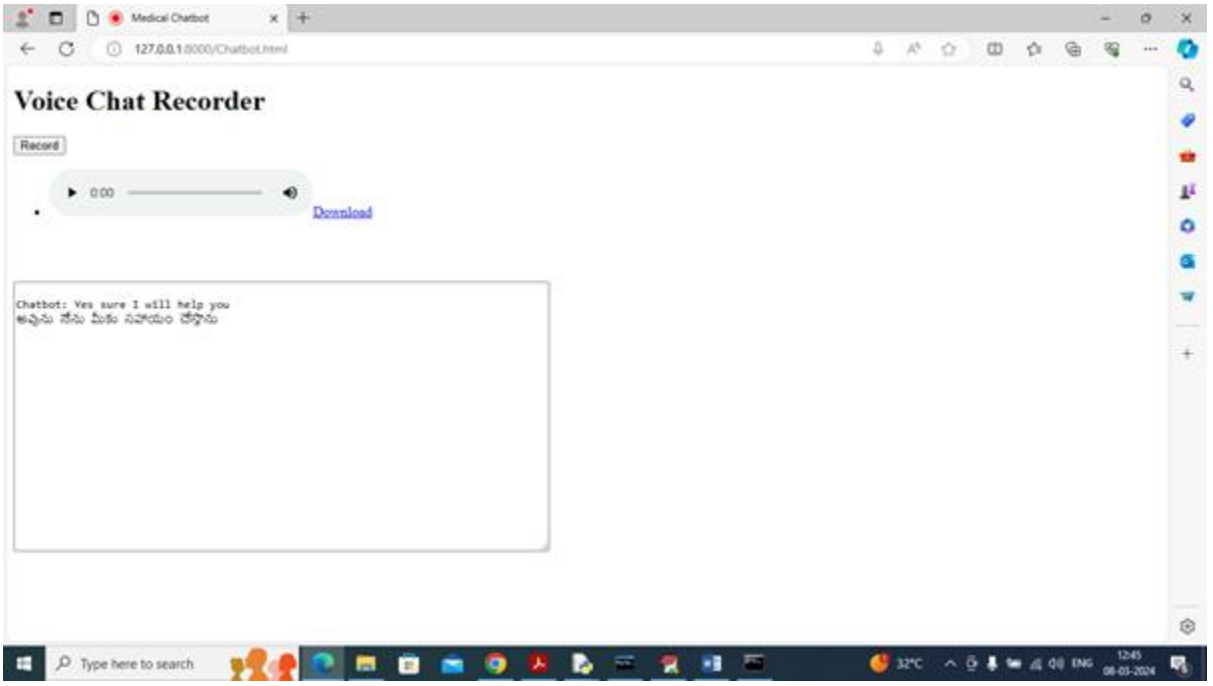
10.2.8 In above screen LSTM training completed and in blue colour text can see LSTM accuracy is 99% and in graph x-axis represents training EPOCHS and y-axis represents Accuracy/LOSS values and then green line represents Accuracy and red line represents LOSS

and can see with each increasing epoch accuracy got increase and reached closer to 1 and loss got decrease. Now click on ‘Interact with Voice Chatbot’ link to get below voice recorder.

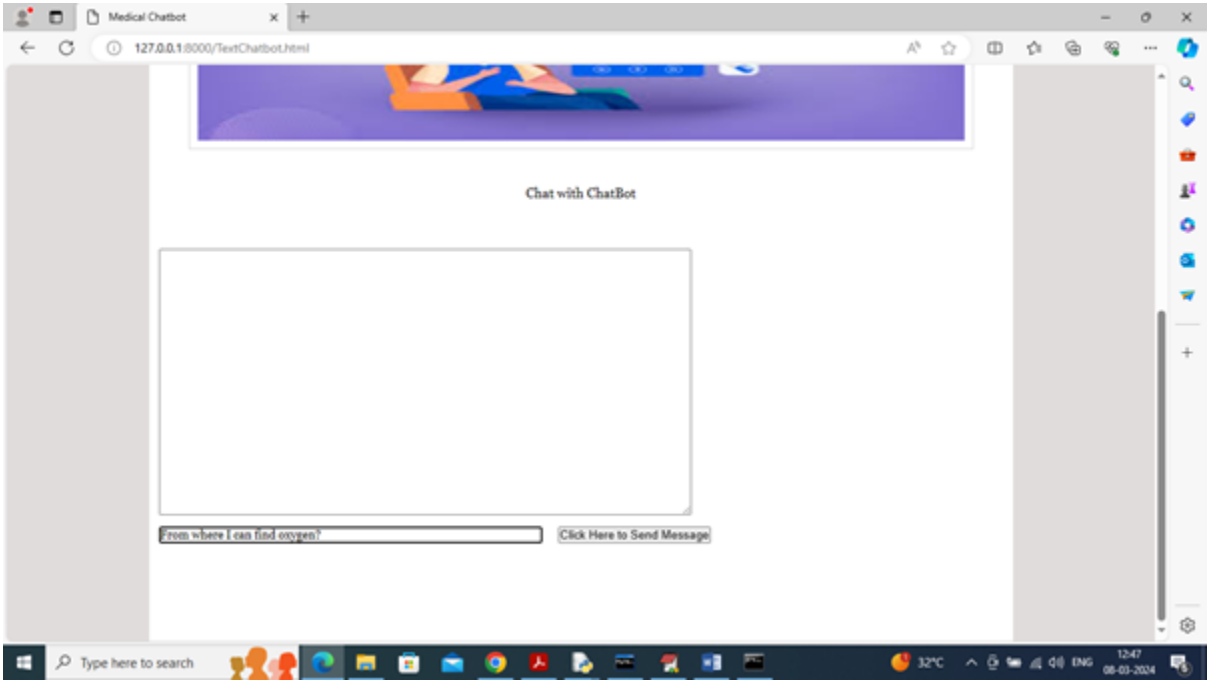
10.2.9 In above screen click on ‘Get Microphone’ link to connect to micro phone and get below page



10.2.10 In above screen click on ‘Record’ button and start speaking and once done click ‘Stop’ button to get reply from Chatbot.



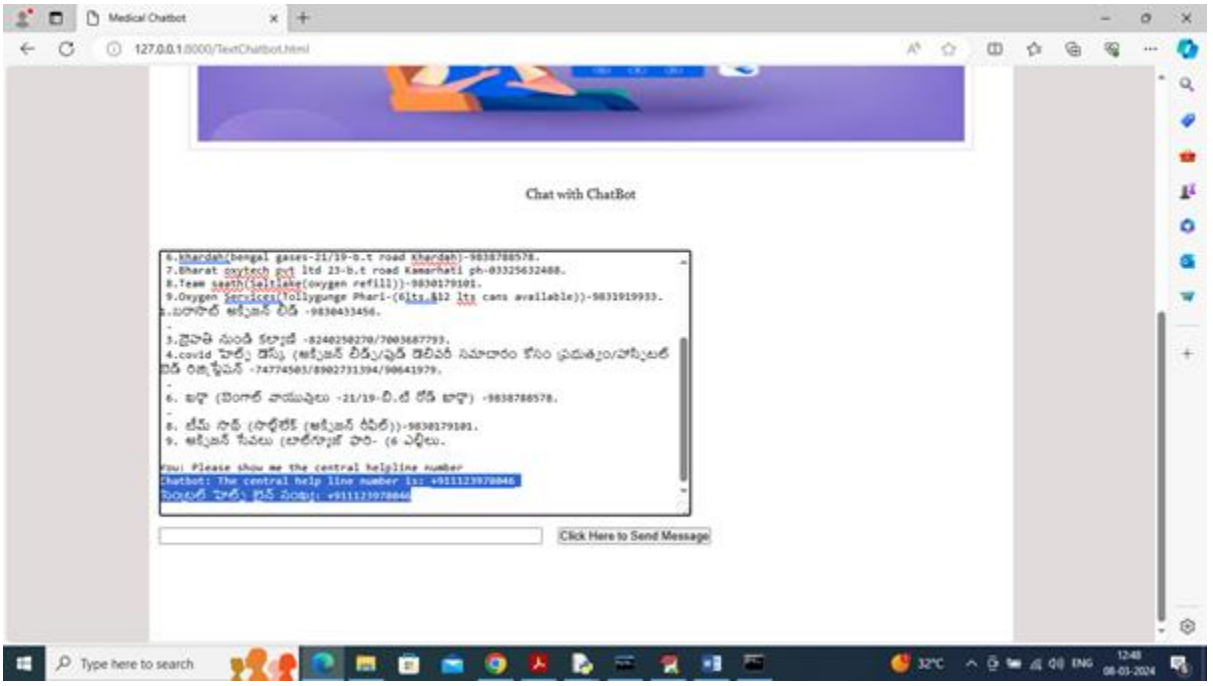
10.2.11 In above screen I spoke word as 'Need Your Help' and then got reply from Chatbot in both English and Telugu and similarly you can record and get output from Chatbot and now click on 'Text Based Chatbot' to get below page.



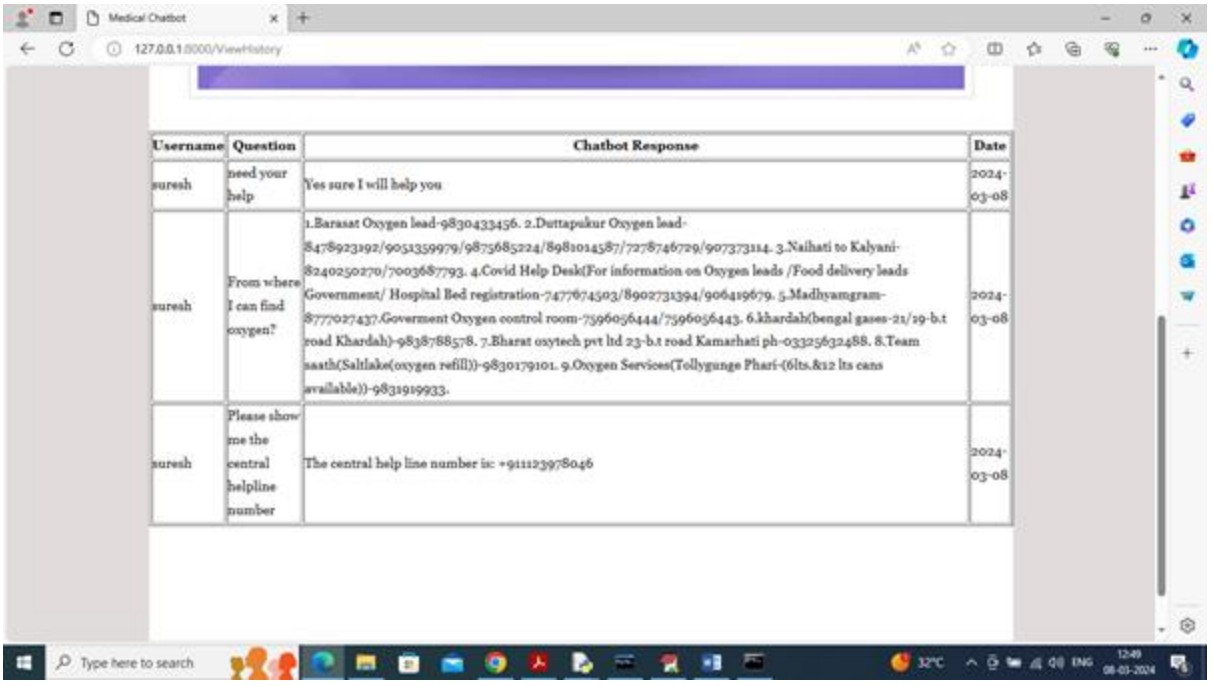
10.2.12 In above screen I asked question about 'Oxygen Cylinder' and press button to get below page.

10.2.13 In above screen got reply from Chatbot in both English and Telugu and below is another question

10.2.14 In above screen asking for ‘covid help line number’ and below is the response



10.2.15 In above screen can see response for help line in both Telugu and English and now click on ‘View History’ link to get below page



Username	Question	Chatbot Response	Date
suresh	need your help	Yes sure I will help you	2024-03-08
suresh	From where I can find oxygen?	1.Barasat Oxygen lead-9830433456. 2.Duttapukur Oxygen lead-8478923192/9051359979/9875685224/8981014587/7278746729/9073731l4. 3.Naihati to Kalyani-8240250270/7003687793. 4.Covid Help Desk(For information on Oxygen leads /Food delivery leads Government/ Hospital Bed registration-7477674503/8902731394/9064199679. 5.Madhyamgram-8777027437.Goverment Oxygen control room-7596056444/7596056443. 6.khardah(bengal gases-21/19-b.t road Khardah)-9838788578. 7.Bharat oxytech pvt ltd 23-b.t road Kamarhati ph-03325632488. 8.Team saath(Saltlake(oxygen refill))-9830179101. 9.Oxygen Services(Tollygunge Phari-(6lts.&12 lts cans available))-9831919933.	2024-03-08
suresh	Please show me the central helpline number	The central help line number is: +911123978046	2024-03-08

In above screen user can view all question he asked and the response from the Chatbot.

11.CONCLUSION

In conclusion, the development of an AI-based medical chatbot model for infectious disease prediction represents a significant advancement in healthcare technology with the potential to revolutionize disease surveillance, diagnosis, and prevention. Through the integration of artificial intelligence, machine learning, and natural language processing techniques, these chatbots offer scalable, accessible, and personalized solutions for individuals and healthcare providers alike.

By leveraging diverse data sources, including clinical records, epidemiological data, and social media, AI-based chatbots can analyze and interpret complex patterns and trends in infectious disease transmission. This enables early detection of outbreaks, timely risk assessment, and proactive interventions, thereby reducing the burden on healthcare systems and mitigating the spread of infectious diseases.

Moreover, the conversational interface of chatbots enhances user engagement and accessibility, empowering individuals with real-time information, personalized recommendations, and self-management tools. Chatbots serve as virtual assistants, facilitating communication and information exchange between users and healthcare professionals, fostering collaborative decision-making and improving health outcomes.

However, the successful implementation of AI-based medical chatbot models for infectious disease prediction requires careful consideration of technical, ethical, and regulatory considerations. Ensuring data privacy, security, and regulatory compliance is paramount to building trust and safeguarding patient rights. Additionally, ongoing research and evaluation are needed to validate the effectiveness, reliability, and usability of chatbot systems in real-world healthcare settings.

In summary, AI-based medical chatbots hold immense promise as innovative tools for infectious disease prediction and management. By harnessing the power of artificial intelligence and human-machine collaboration, we can enhance our ability to respond to infectious disease threats effectively, protect public health, and improve the well-being of individuals and communities worldwide. As technology continues to evolve, the potential for chatbots to make a positive.

11.1 FUTURESCOPE

The development of an AI-based medical chatbot model for infectious disease prediction is a dynamic and evolving field, with several avenues for future research and innovation. As technology continues to advance and new challenges emerge, there are several key areas that warrant further investigation and development.

1. **Enhanced Predictive Models**: Future work could focus on refining and enhancing the predictive models used in AI-based chatbots for infectious disease prediction. This may involve exploring advanced machine learning techniques, such as deep learning, ensemble methods, and reinforcement learning, to improve the accuracy, robustness, and generalizability of the models. Additionally, integrating multimodal data sources, such as genomic data, environmental sensors, and wearable devices, could further enhance predictive capabilities.

2. **Real-Time Surveillance and Early Warning Systems**: Building real-time surveillance and early warning systems for infectious diseases is crucial for timely detection and response. Future research could explore the development of AI-driven algorithms for continuous monitoring of disease outbreaks, transmission dynamics, and emerging threats. Incorporating spatial-temporal modeling, network analysis, and anomaly detection techniques could enable proactive interventions and resource allocation.

3. **Personalized Risk Assessment and Intervention**: Tailoring risk assessment and intervention strategies to individual characteristics and preferences is essential for personalized healthcare delivery. Future work could focus on integrating personalized medicine principles into AI-based chatbots, including genetic profiling, lifestyle factors, and socio-economic determinants of health. This could enable chatbots to provide more targeted recommendations and support for disease prevention and management.

4. **Interoperability and Integration**: Seamless interoperability and integration with existing healthcare systems and digital health platforms are critical for the adoption and scalability of AI-based chatbots in clinical practice. Future research could explore interoperability standards, data exchange protocols, and integration frameworks to facilitate seamless communication and data sharing between chatbots, electronic health records, telemedicine platforms, and other health information systems.

5. **Ethical and Regulatory Considerations**: Addressing ethical and regulatory considerations is essential for ensuring the responsible and ethical use of AI-based chatbots in healthcare. Future work could focus on developing ethical guidelines, best practices, and governance frameworks for chatbot development, deployment, and evaluation. Additionally, addressing concerns related to data privacy, informed consent, bias mitigation, and algorithmic transparency will be crucial for building trust and acceptance among users and stakeholders.

In conclusion, future research in AI-based medical chatbots for infectious disease prediction should focus on advancing predictive modeling techniques, enhancing real-time

surveillance capabilities, personalizing healthcare interventions, promoting interoperability and integration, and addressing ethical and regulatory challenges. By addressing these key areas, we can unlock the full potential of AI-driven chatbots to transform infectious disease prediction and management, ultimately improving public health outcomes and enhancing the quality of care for individuals worldwide.

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